

A Fixed- k Bipower Jump Test

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Abstract

We develop a jump test for Itô semimartingales observed at high frequency based on a ratio of *untruncated* realized variance to bipower variation within a block of k consecutive returns, where k is held *fixed*. Under the infill asymptotics, the limiting distribution of our proposed test is asymptotically pivotal under the null of no jumps according to a coupling result inherited from the driving Brownian diffusion, and therefore yields favorable statistical properties over short time window. Monte Carlo experiments calibrated to empirical financial data show that the proposed test attains essentially nominal size at small k . The framework supplies a single, tuning-free, block-localized price-jump detector that plugs directly into the wider apparatus of fixed- k volatility inference of [Bollerslev, Li, and Liao \(2021\)](#) and bipower jump test of [Barndorff-Nielsen and Shephard \(2006\)](#).

Keywords: jump test, bipower variation, fixed- k asymptotics, high-frequency data, coupling, pivotal inference.

JEL: C12, C14, C22, C58.

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1 Introduction

News arrivals and asset-price jumps appear intimately connected. As the efficient market hypothesis reveals, information that changes beliefs about future cash flows, discount rates, or risk should be rapidly incorporated into prices. When investors' valuation revisions are directionally aligned, they generate one-sided order flow and induce rapid repricing in a instant as visually sharp price movements ([Fleming and Remolona \(1999\)](#), [Andersen, Bollerslev, Diebold, and Vega \(2003\)](#)), which motivates the jump analysis in modelling efficient prices. Testing jump existence around news arrival times is therefore an important question. This paper develops a pivotal test for jumps with high-frequency data over time periods of direct interest to empirical researchers.

Our test builds on the central insight of the bipower jump tests of [Barndorff-Nielsen and Shephard \(2004, 2006\)](#) (henceforth BNS): untruncated realized variance (RV) and bipower variation (BV) respond differently to jumps. It is noteworthy that RV adds up every squared high-frequency returns, including ones containing jumps, and therefore converges to total quadratic variation with the squared jump contribution left intact. By contrast, BV sums up products of absolute returns from adjacent, nonoverlapping sampling intervals, which eliminate the jump contribution by multiplying jumps with neighboring continuous returns with additional finite-activity jump assumption; BV consequently converges to integrated variance and is jump robust. BNS exploit this RV–BV discrepancy through a linear difference or a ratio. Under the no-jump null, both RV and BV converge to integrated variance. Under a finite-activity jump alternative, RV instead converges to integrated variance plus total jump variation, whereas BV continues to converge to integrated variance. This asymptotic separation is the source of identification that we carry into our fixed- k jump test.

What changes is the experiment used to calibrate the discrepancy. For a block j of k returns, we invert the BNS ratio and use $T_{n,j} = \hat{c}_{n,j}^{\text{RV}} / \hat{c}_{n,j}^{\text{BV}}$, so jump evidence appears in its upper tail, while k remains fixed as the sampling mesh shrinks. The fixed- k coupling of [Bollerslev, Li, and Liao \(2021\)](#) approximates the return vector on a continuous block by k iid Gaussian shocks multiplied by a common spot-volatility scale. Since RV and BV are homogeneous of the same degree, the

scale cancels from their ratio. The raw ratio therefore does not collapse to one under the null, as in large-sample BNS theory, but converges to a nondegenerate pivotal functional $T_k^\star$ whose critical values depend only on k . Under an isolated fixed-magnitude jump J , RV receives a contribution of order J^2/Δ_n , whereas BV receives only order $|J|/\sqrt{\Delta_n}$ through products with adjacent continuous returns. Hence $T_{n,j}$ diverges at rate $\Delta_n^{-1/2}$. The test preserves the BNS jump signal while replacing its within-window law of large numbers, Gaussian studentization, and quarticity estimation with a finite-dimensional reference distribution.

This representation also reveals a deeper connection with [Dovonon, Gonçalves, Hounyo, and Meddahi \(2019\)](#), henceforth DGHM. DGHM generate, under the null, conditionally Gaussian pseudo-returns with locally estimated variances and bootstrap a recentered and studentized RV–BV statistic. In effect, they estimate and resample a local-Gaussian backbone. The BLL coupling exposes the same backbone directly in the original returns over a shrinking fixed- k block, where volatility becomes a common local scale; our ratio then removes that scale. What serves as a conditional resampling experiment in DGHM therefore becomes the limiting reference experiment here. The statistics are different—DGHM refine a many-return studentized difference, whereas we calibrate a finite-dimensional ratio—but the underlying Gaussian idea is the same. Once k and the event blocks are specified, critical values can be simulated once and tabulated, without local-volatility estimation, truncation, recentering, or a bootstrap loop. In this sense, BNS supplies the jump-sensitive contrast, BLL its fixed-dimensional approximation, and DGHM its bootstrap interpretation.

Finally, a prespecified event window may contain several candidate blocks, turning window-level jump detection into a multiple-testing problem. For a fixed collection $\mathcal{M}$ of disjoint blocks, the statistics converge jointly under the no-jump null to iid copies of $T_k^\star$: the Gaussian shocks are independent across blocks, and volatility scales cancel block by block. A maximum test with a Šidák critical value ([Šidák, 1968](#)) therefore controls the family-wise error rate, while its maximizing block records where the evidence occurs. We also obtain conservative uniform size control for a growing scan under additional conditions. The procedure thus delivers a window-level decision

without discarding event-time localization.

Our work also relates to the broader literature on high-frequency jump detection and event-time inference. Beyond the RV–BV approach of BNS, jump tests have been constructed from power variations sampled at different frequencies (Aït-Sahalia and Jacod, 2009), locally standardized extreme returns (Lee and Mykland, 2008), and truncated power variations (Podolskij and Ziggel, 2010); Maneesoonthorn, Martin, and Forbes (2020) provide a comparative assessment. Our shrinking-block formulation is complementary to permutation-based tests for distributional discontinuities at known event times (Bugni, Li, and Li, 2023). Its event-study motivation is supported by evidence linking intraday jumps to public announcements (Lahaye, Laurent, and Neely, 2011; Bajgrowicz, Scaillet, and Treccani, 2016; Aït-Sahalia, Li, and Li, 2025).

The remainder of the paper is organized as follows. Section 2 states the semimartingale setting and the local Gaussian approximation. Section 3 defines the fixed- k bipower statistic and develops the block-level theory. Section 4 gives the multi-block extensions. Section 5 studies local power. Section 6 presents the Monte Carlo evidence. Section 7 presents the event-study application. Section 8 concludes. Proofs, additional simulations and empirical checks, and critical-value tables are collected in the appendices.

2 Setup and Local Gaussian Approximation

Motivated by no arbitrage rationale (Delbaen and Schachermayer (1994)), we assume the univariate log-price process $X = (X_t)_{t \geq 0}$ to be an Itô semimartingale defined on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$, written as

$$X_t = X_0 + \int_0^t b_s \, ds + \int_0^t \sigma_s \, dW_s + J_t, \quad (1)$$

where W is a standard Brownian motion driving X , the drift b and volatility σ are càdlàg adapted processes, and J is a pure-jump process with locally bounded compensator. We denote the spot variance by $c_t \equiv \sigma_t^2$. In a finite fixed time horizon $[0, T]$, the econometrician discretely observes X

based on a regular time grid $0, \Delta_n, 2\Delta_n, \dots, n\Delta_n$ with $\Delta_n \equiv T/n$ and records the n intraday returns. The log-return over i -th sampling interval is denoted by $\Delta_i^n X$ and defined as

$$\Delta_i^n X \equiv X_{i\Delta_n} - X_{(i-1)\Delta_n}, \quad i = 1, \dots, n.$$

Throughout the paper, we normalize $T = 1$ (one trading day) without loss of generality. Under the fixed- k setting, we slice each trading day into m_n non-overlapping blocks of exactly k consecutive returns. And we adopt the following notations for class of log-return indices and block time intervals:

$$\mathcal{I}_{n,j} \equiv \{(j-1)k + 1, \dots, jk\}, \quad \mathcal{T}_{n,j} \equiv [(j-1)k\Delta_n, jk\Delta_n],$$

for $j = 1, \dots, m_n$, with $m_n = \lfloor n/k \rfloor$. Let $t(n, j) \equiv (j-1)k\Delta_n$ denote the starting time of block j . Given the sample n , for each j , we conduct blockwise jump testing with the null and alternative specified as follows.

$$\begin{aligned} H_0^j : \omega \in \Omega_{0,j} &\equiv \{\omega \in \Omega : t \mapsto X_t(\omega) \text{ is continuous on } \mathcal{T}_{n,j}\} \\ H_1^j : \omega \in \Omega_{1,j} &\equiv \{\omega \in \Omega : t \mapsto X_t(\omega) \text{ is discontinuous on } \mathcal{T}_{n,j}\} \end{aligned} \tag{2}$$

To derive the limit theory under which k is held fixed in infill asymptotics, we adopt the following very mild conditions as the fixed- k theory of [Bollerslev, Li, and Liao \(2021\)](#).

Assumption 1. Suppose that X has the specification (1) and there exist a sequence of stopping times $(T_m)_{m \geq 1}$ tending to infinity and a bounded sequence of constants $(K_m)_{m \geq 1}$ satisfying the following conditions for every $m \geq 1$: (i) for some constant $r \in [0, 2]$, $|b_t| + |\sigma_t| + |\sigma_t^{-1}| + \int (|x|^r \wedge 1) F_t(dx) \leq K_m$ for all $t \in [0, T_m]$, where $F_t(\cdot)$ is the spot Lévy measure of the jump process J ; (ii) for some $\varkappa > 0$, $\mathbb{E}[|\sigma_{t \wedge T_m} - \sigma_{s \wedge T_m}|^2] \leq K_m |t - s|^{2\varkappa}$ for all $s, t \in [0, T]$.

The above regularization conditions guarantee that each component of process X is locally well-behaved: condition (i) ensures that drift is bounded, the volatility is bounded away from above and below, the activity index r governs the intensity of small jumps: when $r = 0$, the jump process is finitely active, a larger r means jumps are more frequent over a given period, and the case $r = 2$ corresponds to the weakest assumption on jumps; condition (ii) implies that the volatility is locally

$\varkappa$ -Hölder continuous, and it is noteworthy that Itô semimartingale, rough and long memory volatility models satisfy this condition¹. We then have the following coupling result.

Lemma 1 (Local Gaussian shift coupling). *Under Assumption 1, for each fixed k and each block j , there exists a coupling of the k -vector of returns on $\mathcal{I}_{n,j}$ with an iid standard normal vector $(Z_i)_{i \in \mathcal{I}_{n,j}}$ and k random variables $(\tilde{J}_i)_{i \in \mathcal{I}_{n,j}}$ such that*

$$\max_{i \in \mathcal{I}_{n,j}} \left| \frac{\Delta_i^n X}{\Delta_n^{1/2}} - \left(\sqrt{c_{t(n,j)}} Z_i + \frac{\tilde{J}_i}{\Delta_n^{1/2}} \right) \right| = O_{\mathbb{P}} \left(\Delta_n^{\varkappa \wedge (1/2)} \right) = o_{\mathbb{P}}(1), \quad \text{for all } 1 \leq j \leq m_n. \quad (3)$$

where $t(n, j) \equiv (j-1)k\Delta_n$. In particular, on the event $\Omega_{0,j}$ defined in (2), we have $\tilde{J}_i = 0$ for all $i \in \mathcal{I}_{n,j}$.

If we treat jumps as nuisance parameters, then the above coupling denote a location shift from a centered Gaussian variable scaled by left-endpoint volatility. Lemma 1 is the entire engine of what follows. Any smooth functional of the block returns that is scale-invariant in $c_{t(n,j)}$ will converge in distribution to the same functional applied to an iid $N(0, 1)$ vector, with no nuisance parameters left behind.

3 The Fixed- k Bipower Jump Test Statistic

For the j -th block, we define its blockwise realized variance and bipower variation estimators of the spot variance $c_{t(n,j)}$:

$$\hat{c}_{n,j}^{\text{RV}} \equiv \frac{1}{k\Delta_n} \sum_{i \in \mathcal{I}_{n,j}} (\Delta_i^n X)^2, \quad \hat{c}_{n,j}^{\text{BV}} \equiv \frac{\mu_1^{-2}}{(k-1)\Delta_n} \sum_{i=(j-1)k+2}^{jk} |\Delta_{i-1}^n X| |\Delta_i^n X|,$$

¹Within the framework of fractional Brownian motion modeling, the literature has documented both long memory (Comte and Renault (1998)) and roughness (Gatheral, Jaisson, and Rosenbaum (2018)) in empirical studies of stochastic volatility models for high-frequency data. This coexistence of empirical evidences reflects the weak identification issue inherent in fractional modeling; see Li, Phillips, Shi, and Yu (2025).

where $\mu_1 \equiv \sqrt{2/\pi} \approx 0.7979$ which is the first absolute moment of $N(0, 1)$. The blockwise fixed- k bipower jump test statistic is defined as follows.

$$T_{n,j} \equiv \frac{\hat{c}_{n,j}^{\text{RV}}}{\hat{c}_{n,j}^{\text{BV}}} = \frac{\mu_1^2(k-1)}{k} \cdot \frac{\sum_{i \in \mathcal{I}_{n,j}} (\Delta_i^n X)^2}{\sum_{i=(j-1)k+2}^{jk} |\Delta_{i-1}^n X| |\Delta_i^n X|}. \quad (4)$$

It should be well noted that the numerator $\hat{c}_{n,j}^{\text{RV}}$ is an *untruncated* version of realized returns thus affected by jumps if exist, and jump robustness will come from the “bipower” denominator directly. Therefore, this ratio design should detect jumps and eliminate the effect of spot diffusion coefficients to achieve the pivotality of its null limit. We reject the null hypothesis “no jump on block j ” when $T_{n,j}$ is large based on critical values which is obtained via numerical simulation of the pivotal limit distribution.

3.1 Pivotal Limit under the Null

To describe the asymptotic distribution of $T_{n,j}$ under the null, let $Y_1, \dots, Y_k \stackrel{\text{iid}}{\sim} N(0, 1)$, then the limiting distribution of $T_{n,j}$ is

$$T_k^\star \equiv \frac{\mu_1^2(k-1)}{k} \cdot \frac{\sum_{i=1}^k Y_i^2}{\sum_{i=2}^k |Y_{i-1}| |Y_i|}, \quad \text{where } \mu_1 = \sqrt{2/\pi}. \quad (5)$$

Let $q_k^\star(p)$ denote the quantile function of $T_k^\star$ for $p \in [0, 1]$. Both $T_k^\star$ and its quantiles depend only on k ; in particular, they are free of any parameters of the data-generating process.

Theorem 1 (Pivotal fixed- k null limit). *Let Assumption 1 hold and suppose X is continuous on $\mathcal{T}_{n,j}$, that is under the null $\Omega_{0,j}$, then $T_{n,j} \xrightarrow{d} T_k^\star$ as $\Delta_n \rightarrow 0$ with k being fixed.*

Figure 1 plots the limit distribution for block sizes $k = 5, 8, 10, 15, 20, 30$. The distribution is right-skewed and bounded away from zero; by Proposition 2, its support is $[\mu_1^2(k-1)/(k \cos\{\pi/(k+1)\}), \infty)$. As k increases, it concentrates around one. Figure 2 plots the upper quantiles $q_k^\star(p)$ of $T_k^\star$ for $p = 0.50, 0.90, 0.95, 0.99$ as a function of k ; they decay monotonically toward one. Figure 3 shows the survival function $\mathbb{P}(T_k^\star > x)$ on a semi-log scale; the upper tail decays at a rate that is

approximately exponential for $k \geq 10$ and polynomial-plus-log for small k . For practical use we tabulate the upper quantiles $q_k^*(0.90)$, $q_k^*(0.95)$, $q_k^*(0.99)$ in Table 1. A fuller quantile table for $k \in \{4, \dots, 50\}$ is in Appendix B.

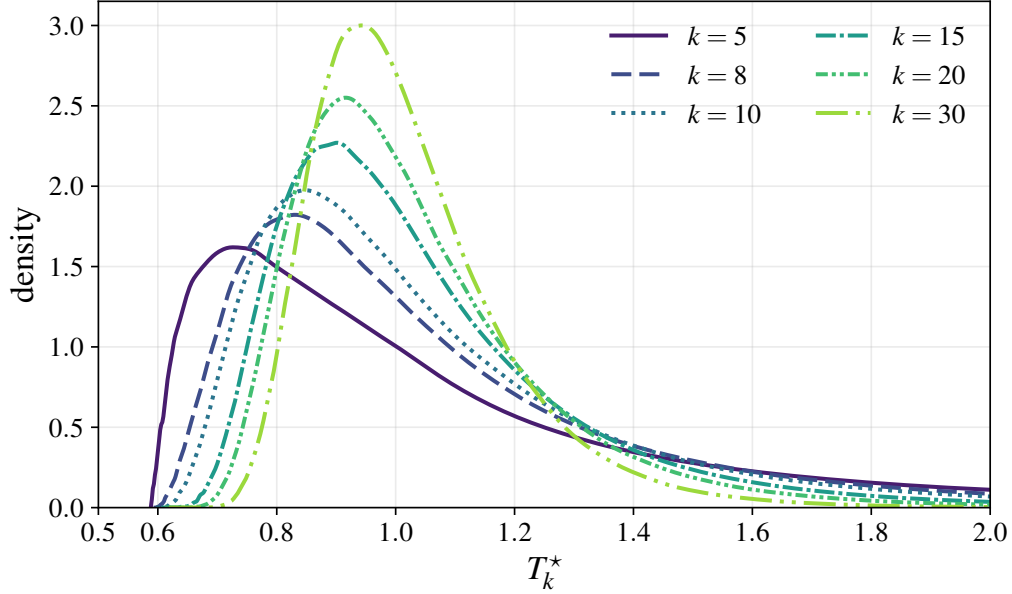


Figure 1: The density of the null limit, T_k^* , for block sizes $k \in \{5, 8, 10, 15, 20, 30\}$. Curves are boundary-aware kernel smoothing estimates from 500,000 Monte Carlo trials, obtained by Gaussian kernel with Silverman bandwidth on the log-transformed sample $\log(T_k^* - L_k)$, where $L_k = \mu_1^2(k-1)/[k \cos(\pi/(k+1))]$ is the sharp support lower endpoint of Proposition 2.

Table 1: Upper quantiles of the fixed- k null distribution T_k^* .

k	LB	med.	$q_k^*(.90)$	$q_k^*(.95)$	$q_k^*(.99)$	k	LB	med.	$q_k^*(.90)$	$q_k^*(.95)$	$q_k^*(.99)$
4	0.590	0.958	2.421	3.606	8.919	15	0.606	0.986	1.379	1.565	2.059
5	0.588	0.970	2.052	2.862	6.243	20	0.612	0.989	1.309	1.448	1.800
6	0.589	0.972	1.860	2.447	4.586	30	0.619	0.992	1.234	1.331	1.564
8	0.593	0.978	1.639	2.021	3.249	40	0.623	0.994	1.196	1.272	1.448
10	0.597	0.980	1.525	1.813	2.671	50	0.625	0.995	1.170	1.235	1.382

Note: Quantiles at the 90%, 95%, and 99% levels for selected block sizes, based on 500,000 Monte Carlo trials. “LB” is the sharp support endpoint $L_k = \mu_1^2(k-1)/[k \cos(\pi/(k+1))]$ from Proposition 2; “med.” is the median of T_k^* . A fuller table for $k \in \{4, \dots, 50\}$ is in Appendix B.

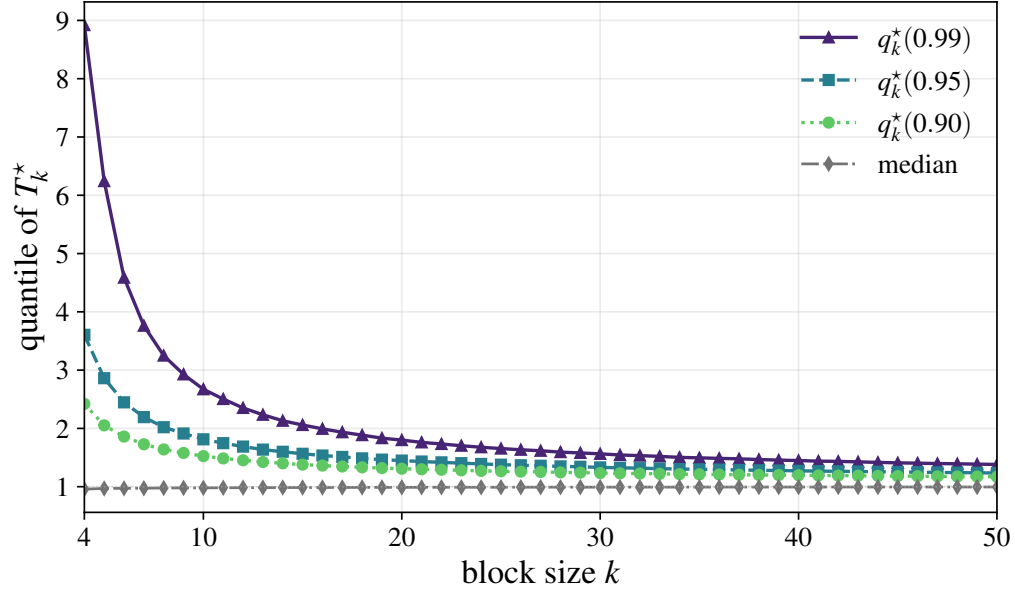


Figure 2: Upper quantiles $q_k^*(p)$ of T_k^* for $p \in \{0.99, 0.95, 0.90, 0.50\}$ as a function of the block size $k \in \{4, \dots, 50\}$. The median is essentially pinned to one for $k \geq 6$.

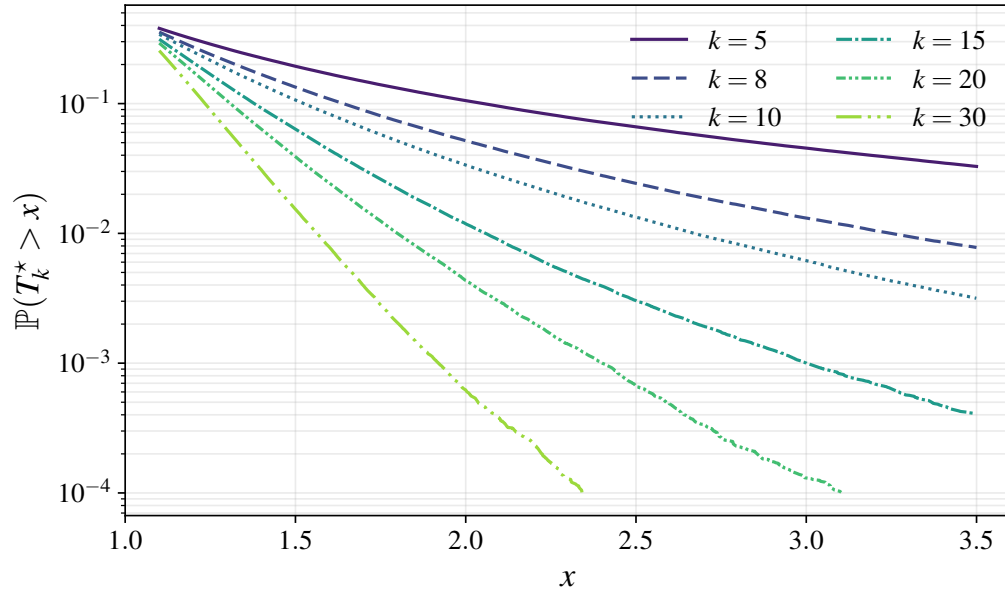


Figure 3: Upper-tail survival function $x \mapsto \mathbb{P}(T_k^* > x)$ on a semi-log scale, for $k \in \{5, 8, 10, 15, 20, 30\}$. The decay is exponential-like for moderate and large k , slower for very small k . This panel is the input into the extreme-value analysis of Section 4.2.

3.2 Consistency under Fixed-Magnitude Jumps

In this subsection, we show that the fixed- k bipower jump test on a single block is consistent against the alternative in which a price jump of fixed magnitude lands inside the block.

Proposition 1 (Divergence under a block-level jump). *Let Assumption 1 hold. Suppose that there is a single jump $\Delta X_{\tau_j} \equiv J \neq 0$, a.s., inside the block $\mathcal{T}_{n,j}$, of random and $O_{\mathbb{P}}(1)$ magnitude. Let $i^* \in \mathcal{I}_{n,j}$ denote the index of the sampling interval containing τ_j , and assume $i^* \notin \{(j-1)k+1, jk\}$. Let $(Z_i)_{i \in \mathcal{I}_{n,j}} \stackrel{iid}{\sim} N(0, 1)$ be the Gaussian couplings part in the Lemma 1, independent of $(c_{t(n,j)}, J)$. Then, as $\Delta_n \downarrow 0$ with k fixed,*

$$T_{n,j} = \Delta_n^{-1/2} \cdot \frac{(k-1) \mu_1^2 |J|}{k \sqrt{c_{t(n,j)}} (|Z_{i^*-1}| + |Z_{i^*+1}|)} \cdot (1 + o_{\mathbb{P}}(1)) \xrightarrow{\mathbb{P}} \infty.$$

Equivalently, $\Delta_n^{1/2} T_{n,j}$ converges in distribution to $(k-1) \mu_1^2 |J| / [k \sqrt{c_{t(n,j)}} (|Y_1| + |Y_2|)]$, with $Y_1, Y_2 \stackrel{iid}{\sim} N(0, 1)$ independent of $(c_{t(n,j)}, J)$.

Proposition 1 shows that $T_{n,j}$ diverges at the canonical BNS rate $\Delta_n^{-1/2}$ under a fixed-magnitude jump inside the block, *without* requiring any quarticity estimator or truncation tuning parameter. The limit distribution of $\Delta_n^{1/2} T_{n,j}$ has a clean closed form (ratio of $|J|$ to a sum of half-normals), which is why power can be traded off against k analytically.

Remark 1 (Multiple jumps in a block). If the block contains $\ell \geq 2$ non-consecutive jumps of magnitudes $J_1, \dots, J_\ell$ at distinct returns, the numerator of $T_{n,j}$ receives $\sum_{\ell'=1}^{\ell} J_{\ell'}^2$ as leading-order contribution, while the denominator receives at most 2ℓ cross terms of order $\Delta_n^{1/2}$. The ratio still diverges at rate $\Delta_n^{-1/2}$. Hence multiple-jump blocks reject at least as strongly as single-jump blocks with the largest $|J|$, and statistical power is weakly monotone in the number of jumps.

Remark 2 (Jumps at block boundaries). If a jump falls at the first return $i^* = (j-1)k+1$ of block j , it shows up unchanged in $\hat{c}_{n,j}^{\text{RV}}$ but in *only one* cross product of $\hat{c}_{n,j}^{\text{BV}}$ instead of two. The test remains consistent against the alternative at rate $\Delta_n^{-1/2}$, with the statement of Proposition 1 adjusted: the two folded-normal sum $|Z_{i^*-1}| + |Z_{i^*+1}|$ in the denominator is replaced by the single folded normal

$|Z_{i^*+1}|$, giving

$$\Delta_n^{1/2} T_{n,j} \xrightarrow{d} \frac{(k-1)\mu_1^2 |J|}{k \sqrt{c_{t(n,j)}} |Z_{i^*+1}|}.$$

The other boundary $i^* = jk$ is symmetric, with $|Z_{i^*-1}|$ in place of $|Z_{i^*+1}|$. For studying jumps triggered by news, if instead the announcement time falls exactly on an even multiple of $k\Delta_n$, the user can shift the block grid by half a block to avoid the boundary problem in event-study design practices for the announcement time is typically known.

4 Multi-Block Extensions

4.1 Fixed Collection of Blocks and Šidák Correction

In many applications (event studies around FOMC, earnings, macroeconomic announcements) the researcher has a pre-specified collection of candidate blocks $\mathcal{M} \subseteq \{1, \dots, m_n\}$ with $|\mathcal{M}| = m$ fixed, and wishes to test the *joint* null $\cap_{j \in \mathcal{M}} \Omega_{0,j}$, i.e., “no jump anywhere in $\cup_{j \in \mathcal{M}} \mathcal{T}_{n,j}$ ”. The natural test statistic is the block maximum

$$T_n^{\max, \mathcal{M}} \equiv \max_{j \in \mathcal{M}} T_{n,j}.$$

Because the blocks are disjoint in time and the Gaussian coupling of Lemma 1 is local within each block, the statistics $(T_{n,j})_{j \in \mathcal{M}}$ are asymptotically *independent*. Consequently, the joint test procedure can be obtained by applying the Šidák’s correction ([Šidák \(1968\)](#)) for the critical value. This is the content of the next theorem.

Theorem 2 (Fixed- m joint limit and Šidák correction). *Let Assumption 1 hold. Under the joint null $\cap_{j \in \mathcal{M}} \Omega_{0,j}$, i.e., X is continuous on $\cup_{j \in \mathcal{M}} \mathcal{T}_{n,j}$, as $\Delta_n \rightarrow 0$ with m and k fixed,*

$$(T_{n,j})_{j \in \mathcal{M}} \xrightarrow{d} (T_k^{\star, (j)})_{j \in \mathcal{M}},$$

where $\{T_k^{\star,(j)}\}_{j \in \mathcal{M}}$ are iid copies of $T_k^{\star}$. Consequently,

$$\mathbb{P}(T_n^{\max, \mathcal{M}} \leq x) \rightarrow F_{T_k^{\star}}(x)^m,$$

in which $F_{T_k^{\star}}$ is the cumulative distribution function of $T_k^{\star}$, and the Šidák-corrected level- α critical value is $\text{cv}_{\alpha, k, m} = q_k^{\star}((1 - \alpha)^{1/m})$.

4.2 Growing m_n and Extreme-Value Asymptotics

When the researcher wishes to detect jumps anywhere in $[0, 1]$ and not at prespecified event blocks, $m_n = \lfloor n/k \rfloor$ grows with n . A Šidák-style critical value remains valid in the sense that the bound

$$\mathbb{P}(T_n^{\max} > q_k^{\star}((1 - \alpha)^{1/m_n})) \leq \alpha + o(1)$$

holds uniformly under the null, see Theorem 3 below. Sharper extreme-value refinements are conceivable along classical lines, but we do not pursue them here: the Šidák bound is already tight enough at the intraday frequencies of interest, for example, $m_n = 130$ for $k = 6$, $n = 780$.

Assumption 2. The Hölder condition Assumption 1(ii) extends to all moments with the same exponent κ : for every integer $p \geq 2$ there exists a finite constant $K_{m,p}$ such that

$$\mathbb{E}[|\sigma_{s \wedge T_m} - \sigma_{t \wedge T_m}|^p] \leq K_{m,p} |s - t|^{p\kappa} \quad \text{for all } s, t \in [0, T], \quad m \geq 1.$$

Theorem 3 (Uniform null bound for $T_n^{\max}$). *Let Assumptions 1 and 2 hold. Set $T_n^{\max} \equiv \max_{j \leq m_n} T_{n,j}$ where $m_n = \lfloor n/k \rfloor$. Under the joint null $\Omega_0 \equiv \cap_{j=1}^{m_n} \Omega_{0,j}$, i.e., X is continuous on $[0, 1]$, for every k satisfying $k \geq 2 \left\lfloor \frac{1}{\kappa \wedge (1/2)} \right\rfloor + 2$ and every significance level $\alpha \in (0, 1)$, we have*

$$\limsup_{n \rightarrow \infty} \mathbb{P}(T_n^{\max} > q_k^{\star}((1 - \alpha)^{1/m_n})) \leq \alpha. \quad (6)$$

where $q_k^{\star}(\cdot)$ is the quantile function of the random variable $T_k^{\star}$.

The three test variants—block-level (Theorem 1), window-level Šidák (Theorem 2), and day-level uniform (Theorem 3)—are now in hand. Algorithm 1 below summarizes the procedure for

empiricists.

Algorithm 1. The fixed- k bipower jump test.

Inputs. Intraday log returns $\{\Delta_i^n X\}_{i=1}^n$ over $[0, 1]$, block size k (default: $k = 10$), nominal level α (default: $\alpha = 0.05$), quantile function $q_k^\star(p)$, see Appendix D.

Steps.

- (1) Partition the return sequence into $m_n = \lfloor n/k \rfloor$ non-overlapping blocks of exactly k consecutive returns.
- (2) For each block $j = 1, \dots, m_n$, compute $T_{n,j}$ from equation (4).
- (3) *Block-level test* (by Theorem 1). Reject “no jump on block j ” at level α whenever $T_{n,j} > q_k^\star(1 - \alpha)$.
- (4) *Window-level test* (by Theorem 2). For a pre-specified window $\mathcal{M}$ of m blocks, reject “no jump in $\mathcal{M}$ ” when $\max_{j \in \mathcal{M}} T_{n,j} > q_k^\star((1 - \alpha)^{1/m})$.
- (5) *Day-level test* (by Theorem 3). Reject “no jump in $[0, T]$ ” when $\max_{j \leq m_n} T_{n,j} > q_k^\star((1 - \alpha)^{1/m_n})$.

Outputs. The full sequence $(T_{n,j})_{j=1}^{m_n}$, its maximum, the Šidák-corrected critical value, and a reject/do-not-reject verdict at each of the three levels above.

5 Local Power Analysis

This section asks the power question for which the test is designed: how small a price jump can be detected in the sampling interval in which a scheduled news release arrives, and what is the statistical price of (i) aggregating returns beyond the announcement block and (ii) not knowing where the jump occurs? Following the event-window perspective of [Bollerslev, Li, Li, and Zhang \(2026\)](#), who model an announcement impact as a local $\Delta_n^{1/2}$ -scale displacement inside a shrinking window, we parametrize the announcement jump on the scale of one local Brownian standard deviation. Three results organize the section. First, $\Delta_n^{1/2}$ is a detection *floor*: below this scale, no test whatsoever can

separate the alternative from the null (Remark 3). Second, the fixed- k block test attains the floor with a nondegenerate limiting power function that we characterize and evaluate (Theorem 4, Figure 4), whereas an RV–BV contrast aggregated over w returns pays a factor $w^{1/4}$ in detectable jump size; the BNS window tests, at $w = n$, are blind to Brownian-scale jumps and become nontrivial only at the coarser Pitman scale $\Delta_n^{1/4}$. Third, location ignorance is costly: the day-level scan over all m_n blocks has no power against Brownian-scale alternatives, because its Šidák critical value diverges. In our discrete-return setting, let $j^\star$ denote the block used to test the announcement, let $L_n \in \{1, \dots, k\}$ denote the within-block return containing the news-induced jump, and write

$$r_n^\star \equiv (j^\star - 1)k + L_n$$

for the corresponding return index. The local announcement-jump alternative is

$$H_{1,n}^{\text{ann}}(\gamma) : \quad \Delta_{r_n^\star}^n J = \gamma \sigma_{\eta_n} \Delta_n^{1/2}, \quad \gamma \neq 0,$$

with no other jump in block $j^\star$, where $\eta_n \in ((r_n^\star - 1)\Delta_n, r_n^\star \Delta_n]$ is the jump time. The exact distribution of η_n inside this single sampling interval is immaterial for returns; taking it uniform, as in [Bollerslev, Li, Li, and Zhang \(2026\)](#), is one natural uninformative convention. Scaling by $\sigma_{\eta_n} \Delta_n^{1/2}$ makes γ the size of the announcement shock in local Brownian standard-deviation units and removes volatility from the limiting power function.

Remark 3 (A Brownian-scale detection floor). The scale $\Delta_n^{1/2}$ in $H_{1,n}^{\text{ann}}(\gamma)$ is not a normalization convenience; it is the finest scale at which announcement jumps remain statistically identifiable. By Lemma 1, the standardized block returns converge, jointly under the null and under $H_{1,n}^{\text{ann}}(\gamma)$, to the k -dimensional Gaussian location experiment in which the observation is $Z + \gamma e_\ell$ with $Z \sim N(0, I_k)$. In that limit experiment,

$$\text{TV}(N(\gamma e_\ell, I_k), N(0, I_k)) = 2\Phi(|\gamma|/2) - 1 \longrightarrow 0 \quad \text{as } \gamma \rightarrow 0,$$

where TV denotes the total variation distance and Φ the standard normal distribution function. Hence, along any drifting sequence $\gamma_n \rightarrow 0$ —that is, for jumps of size $o_{\mathbb{P}}(\sigma_{\eta_n} \Delta_n^{1/2})$ —the power

of *every* test in the limit experiment converges to its size, even when the announcement block $j^\star$ and the within-block position ℓ are known exactly. In this sense $\Delta_n^{1/2}$ is the detection floor of the announcement-jump problem, and Theorem 4 below shows that the fixed- k block statistic attains it: the statistic has a nondegenerate limiting power function at the $\Delta_n^{1/2}$ scale while requiring no knowledge of the local volatility, which the ratio removes automatically.²

The comparison with the BNS family is sharpest when framed as a *cost of aggregation*. Consider an RV–BV contrast computed from w consecutive returns that contain a single jump of size J_n . The jump contributes J_n^2 to the discrepancy, the sampling variability of the contrast under the null is of order $\sqrt{w} \Delta_n$ by the usual central limit scaling, and the two adjacent bipower products contaminated by the jump are of the smaller order $|J_n| \Delta_n^{1/2}$ whenever $|J_n| \gg \Delta_n^{1/2}$. A w -return contrast can therefore see the jump only when $J_n^2 \gtrsim \sqrt{w} \Delta_n$, that is, when

$$J_n \gtrsim w^{1/4} \Delta_n^{1/2} :$$

every widening of the window inflates the minimal detectable jump by the factor $w^{1/4}$. The BNS tests sit at the endpoint $w = n$. Their difference statistic is normalized as $\Delta_n^{-1/2} \{\mu_1^{-2} BV_n - RV_n\}$, and the ratio statistics $\hat{H}$ and $\hat{J}$ share its local rate because near the null

$$\frac{\mu_1^{-2} BV_n}{RV_n} - 1 = \frac{\mu_1^{-2} BV_n - RV_n}{IV} + \text{higher-order terms},$$

where $IV = \int_0^T \sigma_s^2 ds$ and the feasible studentisers in $\hat{H}$ and $\hat{J}$ are $O_{\mathbb{P}}(1)$. Under a single shrinking jump $J_n = \eta \Delta_n^a$, the first-order jump signal in $\hat{G}$, $\hat{H}$, and $\hat{J}$ is therefore governed by

$$\Delta_n^{-1/2} J_n^2 = \eta^2 \Delta_n^{2a-1/2},$$

which is nondegenerate only at the Pitman scale $a = 1/4$, in agreement with the $w^{1/4}$ pricing at $w = n$. At the announcement scale $a = 1/2$ the signal vanishes and the BNS tests retain only

²Because the spot variance $c_{t(n,j^\star)}$ is unknown, feasible tests are constrained to be invariant to the block scale, and the relevant optimality benchmark at the boundary is the Neyman–Pearson envelope of the scale-invariant reduction of the shift experiment; [Bollerslev, Li, Li, and Zhang \(2026\)](#) carry out the analogous envelope computation for their candlestick functionals. We do not pursue the envelope calculation here.

trivial local power. The fixed- k statistic sits at the opposite endpoint $w = k = O(1)$, where the aggregation factor is bounded and the floor of Remark 3 is within reach. The intermediate range is not genuinely available to the BNS construction: its within-window law of large numbers and Gaussian studentization require $w \rightarrow \infty$, and Table 4 documents the size failure that emerges when the BNS window is shortened at a fixed sampling frequency. At w fixed the analysis also changes character: at the boundary $J_n \asymp \Delta_n^{1/2}$ the bipower contamination $|J_n| \Delta_n^{1/2}$ is of the *same* order as J_n^2 , so a $\Delta_n^{1/2}$ -scale jump does not enter as a mean shift of an asymptotically Gaussian statistic; it shifts one coordinate of the Gaussian limit experiment, and the resulting limit law is the full functional perturbation described next.

Write the fixed- k ratio functional as

$$\mathcal{R}(y_1, \dots, y_k) \equiv \frac{\mu_1^2(k-1)}{k} \frac{\sum_{i=1}^k y_i^2}{\sum_{i=2}^k |y_{i-1}| |y_i|},$$

so that the null limit is $T_k^\star = \mathcal{R}(Z)$ for $Z \sim N(0, I_k)$.

Theorem 4 (Local power against announcement-time jumps). *Let Assumption 1 hold and let $j^\star = j_n^\star \in \{1, \dots, m_n\}$ be the block used to test the announcement. Under $H_{1,n}^{\text{ann}}(\gamma)$, suppose L_n is deterministic or independent of the Brownian increments in block $j^\star$, and $L_n \xrightarrow{d} L$, where L takes values in $\{1, \dots, k\}$, is independent of $Z \sim N(0, I_k)$, and satisfies $\mathbb{P}(L = \ell) = \pi_\ell$. Define*

$$G_{\gamma,L} \equiv \mathcal{R}(Z + \gamma e_L),$$

where e_L is the corresponding unit vector in $\mathbb{R}^k$, and write F_k for the distribution function of $T_k^\star$.

(i) Single-block test. For every $k \geq 3$ and every fixed $\alpha \in (0, 1)$,

$$T_{n,j^\star} \xrightarrow{d} G_{\gamma,L},$$

and hence

$$\lim_{n \rightarrow \infty} \mathbb{P}_{H_{1,n}^{\text{ann}}(\gamma)}(T_{n,j^\star} > q_k^\star(1 - \alpha)) = \sum_{\ell=1}^k \pi_\ell \mathbb{P}(\mathcal{R}(Z + \gamma e_\ell) > q_k^\star(1 - \alpha)).$$

(ii) Fixed event window. Let $\mathcal{M}_n \subset \{1, \dots, m_n\}$ be a prespecified collection of m blocks, with m fixed and $j^\star \in \mathcal{M}_n$, and suppose all blocks in $\mathcal{M}_n \setminus \{j^\star\}$ are continuous. Then

$$T_n^{\max, \mathcal{M}_n} \xrightarrow{d} \max\{G_{\gamma, L}, T_k^{\star, (1)}, \dots, T_k^{\star, (m-1)}\},$$

where $T_k^{\star, (1)}, \dots, T_k^{\star, (m-1)}$ are iid copies of $T_k^\star$, independent of (Z, L) . With the Šidák critical value $c_{k, m}(\alpha) \equiv q_k^\star((1 - \alpha)^{1/m})$,

$$\lim_{n \rightarrow \infty} \mathbb{P}_{H_{1, n}^{\text{ann}}(\gamma)}(T_n^{\max, \mathcal{M}_n} > c_{k, m}(\alpha)) = 1 - F_k(c_{k, m}(\alpha))^{m-1} \left[\sum_{\ell=1}^k \pi_\ell \mathbb{P}(\mathcal{R}(Z + \gamma e_\ell) \leq c_{k, m}(\alpha)) \right].$$

(iii) Day-level growing maximum. Suppose in addition that Assumption 2 holds, that k satisfies the block-size condition in Theorem 3, and, for simplicity, $n = m_n k$. If a single announcement-time local jump occurs in block $j^\star$ and all other blocks are continuous, then with $c_{k, m_n}(\alpha) \equiv q_k^\star((1 - \alpha)^{1/m_n})$,

$$\mathbb{P}(G_{\gamma, L} > c_{k, m_n}(\alpha)) \rightarrow 0, \quad \limsup_{n \rightarrow \infty} \mathbb{P}_{H_{1, n}^{\text{ann}}(\gamma)}(T_n^{\max} > c_{k, m_n}(\alpha)) \leq \alpha.$$

In the Gaussian reference experiment with iid null blocks, the last display is sharp and the rejection probability tends to α . Thus a Brownian-scale announcement jump has nondegenerate local power for block and fixed-window tests, but not for a day-level maximum whose critical value diverges with m_n .

Because $G_{\gamma, L}$ is a functional of a standard Gaussian vector, the limiting power functions delivered by Theorem 4 are free of all nuisance parameters and can be evaluated once and for all; Figure 4 does so at the 5% level. Three lessons emerge. First, announcement-scale detection is demanding: at $k = 10$, a jump equal to four local standard deviations landing in the interior of the block is detected with probability 25.5%, and even odds require $\gamma \approx 5.7$. The block size acts as expected from Section 6: power at the boundary scale rises with k , with $\gamma_{50\%}$ falling from 7.4 at $k = 6$ to 5.7 at $k = 10$ and 5.2 at $k = 15$, so k buys boundary-scale power at the price of the coupling error documented there. Second, the *position* of the jump inside the block matters, in a direction that is useful for design. A jump on the first return of the block contaminates one bipower cross-product

instead of two, so the denominator of $T_{n,j}$ is less inflated and power is substantially higher: at $\gamma = 4$, first-return power is 49.2% against 25.5% in the interior, and $\gamma_{50\%}$ improves from 5.7 to 4.0. This gives a power argument for the event-study alignment of Section 7, which places the announcement minute at the first observation of its block; that alignment is motivated there by robustness to the post-announcement volatility shift, and the two considerations point to the same design. Third, the Šidák window test inherits most of the block-level power: with $m = 3$ blocks and the jump on the first return of one of them—the exact configuration of the FOMC application—the 5% window test detects a four-standard-deviation jump with probability 31.8% against the oracle single block’s 49.2%, the difference being the price of guarding all three blocks without knowing which one contains the jump.

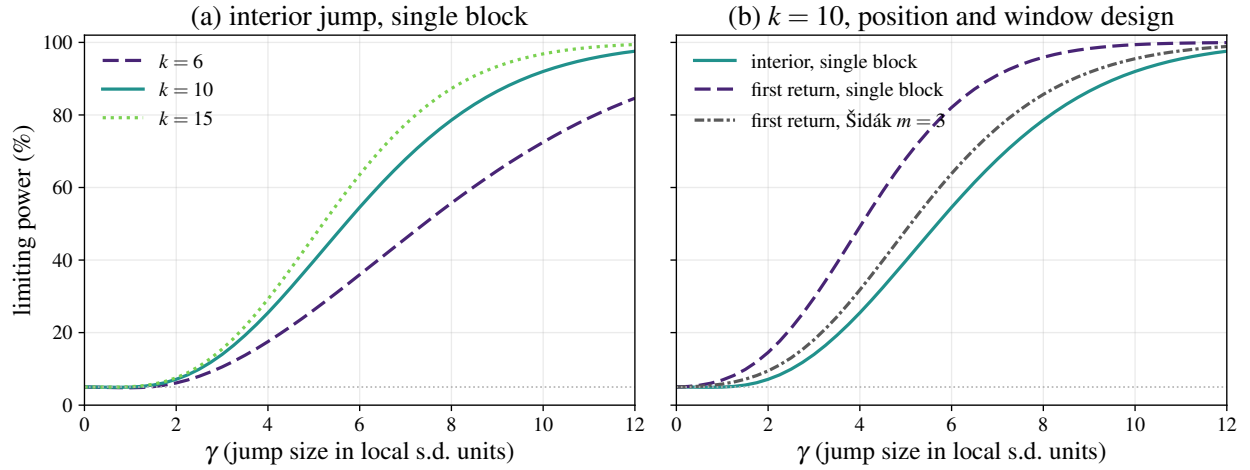


Figure 4: Limiting power of the fixed- k bipower jump test against the announcement-time local alternative $H_{1,n}^{\text{ann}}(\gamma)$, evaluated in the Gaussian limit experiment of Theorem 4 by Monte Carlo (2,000,000 trials per curve, common random numbers across γ), at nominal level $\alpha = 5\%$. Panel (a): single-block test with the jump in the interior of the block, $k \in \{6, 10, 15\}$. Panel (b): $k = 10$, comparing the jump in the interior versus on the first return of the block (one contaminated bipower cross-product instead of two) and the Šidák window test over $m = 3$ blocks with the jump on the first return of one block—the configuration of the FOMC design in Section 7; the Šidák critical value is $q_{10}^*(0.95^{1/3}) = 2.357$. Dotted horizontal lines mark the nominal level.

The three results of this section thus fit a single ordering: the value of knowing where to look. With the announcement block known, the fixed- k test operates at the detection floor $\Delta_n^{1/2}$ of Remark 3. With only the event window known, the Šidák test pays a modest and exactly quantifiable premium, displayed in Figure 4(b). With no location information at all, the day-level scan of Theorem 4(iii) is

powerless against Brownian-scale jumps: its critical value $q_k^*((1-\alpha)^{1/m_n})$ diverges—at a polynomial rate governed by the lower-tail cone geometry established in the proof of Theorem 3, a reflection of the polynomial upper tail of T_k^* —so a bounded shift of one Gaussian coordinate is eventually invisible, and detecting jumps at fully unknown locations requires alternatives that grow with m_n , a regime we leave for future work. The BNS window tests occupy a different point on the same map: agnostic about location *and* aggregated over the full window, they are calibrated to the coarser Pitman scale $\Delta_n^{1/4}$. Appendix C.1 illustrates the resulting rate separation in simulation. Under $J_n = 4\Delta_n^{1/2}$, the fixed- k block and window tests have stable nontrivial power and the fixed- k day-level scan decays toward size, exactly as parts (i)–(iii) predict, while size-adjusted BNS- $\hat{G}$ and BNS- $\hat{J}$ return toward size as n grows. Under $J_n = 0.75\Delta_n^{1/4}$, the BNS tests have nontrivial power because the alternative sits at their Pitman scale, and the fixed- k test gains power as well because the standardized displacement $0.75\Delta_n^{-1/4}$ grows with n : the alternative is above the fixed- k boundary rather than on it.

6 Monte Carlo Simulations

We now assess size and power of our test in a Monte Carlo simulation following [Dovonon, Gonçalves, Hounyo, and Meddahi \(2019, Section 5\)](#), whose data generating process (DGP) is the two-factor stochastic volatility model (SV2F) calibration of [Huang and Tauchen \(2005\)](#). Let the log-price evolve as follows,

$$\begin{aligned}
dX_t &= adt + \sigma_{u,t}\sigma_{sv,t}dW_t + dJ_t, \\
\sigma_{u,t} &= C + A \cdot \exp(-a_1t) + B \cdot \exp(-a_2(1-t)), \\
\sigma_{sv,t} &= s \cdot \exp(\beta_0 + \beta_1\tau_{1,t} + \beta_2\tau_{2,t}), \\
d\tau_{1,t} &= \alpha_1\tau_{1,t}dt + dB_{1,t}, \\
d\tau_{2,t} &= \alpha_2\tau_{2,t}dt + (1 + \phi\tau_{2,t})dB_{2,t}, \\
\text{corr}(dW, dB_1) &= \rho_1, \quad \text{corr}(dW, dB_2) = \rho_2
\end{aligned} \tag{7}$$

with parameters $a = 0.03$, $\beta_0 = -1.2$, $\beta_1 = 0.04$, $\beta_2 = 1.5$, $\alpha_1 = -0.00137$, $\alpha_2 = -1.386$, $\phi = 0.25$, $\rho_1 = \rho_2 = -0.3$. The function s-exp is the spline-exponential of [Huang and Tauchen \(2005\)](#), defined as $\text{s-exp}(x) = \exp(x)$ if $x \leq \log(1.5)$ and $\text{s-exp}(x) = 1.5[\log(1.5)(1 - \log(1.5)) + x^2]^{-1/2}$ otherwise. For the initialization of factors, we set $\tau_{1,0} \sim N(0, -1/(2\alpha_1))$, and $\tau_{2,0} = 0$. The process $\sigma_{u,t}$ captures the diurnal U -shape of intraday volatility pattern with calibration $A = 0.75$, $B = 0.25$, $C = 0.88929198$. By contrast, setting $(A, B, C) = (0, 0, 1)$ removes the diurnal effect from the model. Moreover, the ratio design of our test make it robust to diurnal effects, so we do not bother to correct the U -shape.

Following [Dovonon, Gonçalves, Hounyo, and Meddahi \(2019\)](#) and references therein, under the null Ω_0 , the jump component J is zero; under the alternative Ω_1 , we consider two scenarios for the jump process J . First, J is a finite-activity compound Poisson process with intensity $\lambda = 0.058$ and jump sizes drawn from $N(0, \sigma_{\text{jmp}}^2)$ with $\sigma_{\text{jmp}}^2 = 1.7241$. Due to the finite activity, some simulated paths may contain no jumps. We therefore compute power conditional on the occurrence of at least one realized jump in the sample path. Second, we consider infinite-activity jumps by letting J be a symmetric tempered stable process with Lévy measure $F_t(dx) = \frac{c_1}{x^{1+r}} e^{-c_2|x|} dx$ with $c_2 = 3$, $r = 0.5$ and c_1 calibrated such that jump variation accounts for 10% of quadratic variation³.

We simulate the X -processes according to (7) over $[0, 1]$ by a first-order Euler scheme on a mesh of $1/23,400$, mimicing one second in one trading day of 6.5 hours. The generated process is then regularly observed starting from 0 at frequencies of 8-min, 5-min, 2-min, 1-min, 30-sec, which respectively correspond to sample size $n = 48, 78, 195, 390, 780$. We select the block size $k = 6, 10, 15$ for comparison as too large k should lead to non-negligible coupling error. We set nominal significance level $\alpha = 5\%$, and calculate the empirical rejection probabilities based on 10,000 Monte Carlo trials for each setting. And the test will apply Šidák-correction if testing over multiple blocks. We determine two benchmarks to compare with our proposed test. One is the adjusted ratio test $\hat{J}$ of Eq. (14) in [Barndorff-Nielsen and Shephard \(2006\)](#), denoted by $\text{BNS-}\hat{J}$,⁴ and

³Specifically, $c_1 = 0.1309$ in the “no-diurnal-effect case”, and $c_1 = 0.1273$ in the “with-diurnal-effect case”.

⁴[Barndorff-Nielsen and Shephard \(2006\)](#) proposed three tests, a linear difference test $\hat{G}$, a ratio test $\hat{H}$, and an adjusted ratio test $\hat{J}$. Based on their simulations, $\hat{J}$ has the best size-power performance, and is accordingly chosen to be

the other are the bootstrap tests proposed by [Dovonon, Gonçalves, Hounyo, and Meddahi \(2019\)](#) which is written as DGHM for short.

The comparison between our test and the BNS- $\hat{J}$ on size and power is displayed in Table 2. Several patterns stand out. First, fixed- k dominates BNS- $\hat{J}$ on size performance at all three k 's and sampling frequencies. While both tests' size declining toward nominal in the infill asymptotics, fixed- k simply starts and stays much closer. Second, the diurnal effect inflates $\hat{J}$'s size by 1% to 2.3% for each Δ_n while leaving fixed- k essentially flat, because the per-block ratio $T_{n,j}$ cancels block volatility level including the “U-shape” multiplier $\sigma_{u,t}$. This showcases a structural, not numerical, advantage of our fixed- k test. Third, the block size k traces a size–power tradeoff. For instance, at $\Delta_n=1$ -min with diurnal effects, raising k from 6 to 15 more than doubles infinite-activity power from 20.6% to 43.5% at the cost of size widening from 4.6% to 6.4%. Larger k amplifies the local-Gaussian coupling error, and thus increases the residual size distortion; at the same time, it incorporates more returns within each block, thereby enhancing the test power under the alternative. Fourth, power against the infinite-activity alternative falls 30%–50% below the finite-activity alternative despite the identical 10% jump variation share, because the former smears the same signal across many small jumps thus no dominant return.

A natural concern with Table 2 is that the BNS- $\hat{J}$ benchmark is well known to be size-distorted. Table 3 addresses this by repeating the exercise against the other benchmarks of DGHM's jump tests Boot1 and Boot2 which bootstrap from a linear test, which we call “AT”, of the form “RV - BV” in the same spirit of BNS's $\hat{G}$ test⁵. According to numerical simulation, two patterns emerge. First, DGHM's bootstrap delivers what its theory promises: the asymptotic linear test's severe over-rejection, for example 14.6% and 18.9% when $\Delta_n = 8$ -min, is brought down to 7.1% and 9.3% by Boot1 and further to 5.8% and 7.6% by the second-order Boot2. The fixed- k test approaches this

our benchmark. Moreover, the $\hat{H}$ is also applied in one simulation of Table 4 to reflect the finite-sample issue. For both BNS- $\hat{J}$ and $-\hat{H}$, we did the necessary factoring “ $n/(n-1)$ ” for bipower estimates and “ $n/(n-3)$ ” for quadpower estimates as finite-sample corrections which is inline with the footnote 4 of DGHM.

⁵See Eq.(4), Eq.(6) and Eq.(7) of DGHM for the exact definitions of AT, Boot1, and Boot2 respectively. Per Monte Carlo replication, Boot1 and Boot2 each invoke $B=999$ wild-bootstrap resamples of jump-truncated returns by scaling iid standard normals independent of data with a thresholding-based block-piecewise large- k spot volatility estimator with bandwidth $k_n=\lceil\sqrt{n}\rceil$ and truncation $u_n=2.3\sqrt{BV_n}n^{-0.4}$.

Table 2: Rejection probabilities: fixed- k bipower vs BNS- $\hat{J}$, nominal level $\alpha = 5\%$.

Δ_n	Size: no jumps				n_J	Power: finite-activity jumps				Power: infinite-activity jumps			
	BNS- $\hat{J}$	$k=6$	$k=10$	$k=15$		BNS- $\hat{J}$	$k=6$	$k=10$	$k=15$	BNS- $\hat{J}$	$k=6$	$k=10$	$k=15$
<i>Panel A: SV2F without diurnal effects</i>													
8-min	11.08	7.03	7.96	9.64	565	81.42	58.94	62.65	72.74	28.67	16.35	21.60	26.42
5-min	9.79	5.89	7.46	8.75	535	83.74	62.99	72.34	78.32	32.31	17.49	25.51	30.62
2-min	8.97	5.57	6.27	6.91	560	87.86	67.14	80.36	85.18	37.96	20.45	31.03	36.82
1-min	7.74	4.42	5.76	6.50	554	88.99	63.36	82.85	87.18	44.51	20.20	36.55	44.21
30-sec	7.10	4.36	5.51	5.74	582	93.13	72.16	88.49	91.58	49.35	23.10	42.31	50.39
<i>Panel B: SV2F with diurnal effects</i>													
8-min	13.41	6.53	8.39	10.78	551	81.49	61.89	65.88	75.32	31.05	17.42	21.93	27.41
5-min	12.94	5.94	7.60	9.04	558	84.23	63.98	72.22	81.00	33.55	17.37	25.39	30.52
2-min	10.37	5.40	6.56	6.82	537	86.41	64.43	79.70	85.85	38.23	20.19	31.89	37.99
1-min	8.86	4.58	5.96	6.35	535	90.65	66.17	85.79	88.79	41.97	20.56	36.72	43.53
30-sec	8.07	4.33	5.39	5.91	589	90.66	70.12	87.61	90.83	48.22	22.87	41.43	50.05

Note: numbers are reported as percentage. Panel A sets no intraday seasonality; Panel B contains diurnal effects calibrated so open/close variance is approximately 3/1.5 times mid-day variance. Column n_J reports the number of finite-activity-jumps sample paths on which at least one Poisson jump realised, used as the conditioning denominator for calculating power; power for infinite-activity jumps is unconditional, as every path has jumps almost surely. All results are based on 10,000 Monte Carlo trials.

benchmark with markedly less machinery: notably, $k=6$ is tighter than Boot1 in every comparison and tighter than even the second-order Boot2 in four of five rows in Panel B. Second, on power the bootstrap dominates fixed- k at $k=10$ by 10%–15% against finite-activity jumps at the coarsest sampling, but the gap closes rapidly with Δ_n : at $\Delta_n=30$ -sec in Panel A, $k=10$ rejects 89.6% of finite-activity-jump days against Boot1&2’s 91.2%, and on the infinite-activity alternative the bootstrap’s advantage over $k=10$ never exceeds 5% anywhere in the table. Again, that $k=6$ ’s power is dominated by $k=10$ ’s reflects the size-power tradeoff the tuning parameter governs. Therefore, the case for fixed- k rests on architectural parsimony rather than statistical efficiency. The fixed- k procedure compares the calculated test to the quantile of T_k^* that depends *only* on k and is tabulated once for all applications: no quarticity, no truncation threshold, no bandwidth requirements, no recentering, no bootstrap loop. When conducting the jump test is emphasized to be an empirical tool rather than the object of study, our pivotal inferential method is valid, straightforward, and more useful in practice than any Edgeworth refinement of an inconsistent first-order asymptotic.

Importantly, a key advantage of our fixed- k procedure is that it enables inference on short

Table 3: Rejection probabilities: fixed- k bipower vs DGHM bootstrap tests, nominal level $\alpha = 5\%$.

Δ_n	Size: no jumps					Power: finite-activity jumps						Power: infinite-activity jumps				
	AT	$k=6$	$k=10$	Boot1	Boot2	n_J	AT	$k=6$	$k=10$	Boot1	Boot2	AT	$k=6$	$k=10$	Boot1	Boot2
<i>Panel A: SV2F without diurnal effects</i>																
8-min	14.60	6.11	7.16	7.12	5.80	574	84.15	62.72	63.94	78.92	77.35	33.79	16.50	21.69	24.51	22.65
5-min	13.38	5.85	7.40	7.10	6.22	543	86.37	66.67	74.03	82.69	81.95	35.75	18.10	24.99	28.52	27.00
2-min	10.35	5.77	6.39	6.25	5.54	562	87.72	62.28	77.76	86.30	85.77	40.40	19.91	31.34	35.89	34.95
1-min	9.21	4.74	5.74	6.26	5.73	587	88.25	66.78	83.48	87.22	86.88	45.07	20.76	37.24	41.67	41.12
30-sec	8.04	5.68	6.79	5.95	5.64	567	91.71	71.78	89.59	91.18	91.18	49.78	24.16	42.92	47.09	46.78
<i>Panel B: SV2F with diurnal effects</i>																
8-min	18.86	6.68	8.46	9.30	7.61	556	81.12	58.99	64.03	75.18	73.92	35.61	16.93	21.71	25.16	23.02
5-min	15.42	6.41	7.84	7.93	6.68	614	87.13	63.68	71.50	83.39	82.74	37.47	18.24	25.54	29.61	27.80
2-min	12.78	5.91	6.53	8.42	7.31	548	89.23	70.26	82.12	87.77	87.23	40.00	19.49	31.78	34.94	33.74
1-min	10.35	4.84	5.99	7.04	6.22	565	93.81	72.74	89.03	92.92	92.39	45.03	21.17	37.32	41.15	40.22
30-sec	7.93	5.71	6.13	5.91	5.46	545	93.58	69.91	87.52	92.66	92.66	49.40	25.32	43.26	46.83	46.18

Note: numbers are reported as percentage. Four tests are reported based on 10,000 trials: “AT” is the linear test of DGHM’s Eq(4), which is in the same spirit of BNS’s difference test $\hat{G}$ with with standard error estimated by tripower variation to the power of $4/3$; the column “ $k=10$ ” is the proposed test with $k = 10$ guided by size-power tradeoff in Table 2; “Boot1” is the local-Gaussian wild bootstrap of DGHM’s Eq(6); “Boot2” is the second-order refined bootstrap test, see DGHM’s Eq (7). Bootstrap calibration follows DGHM’s design: $B = 999$ wild-bootstrap resamples per replication and all other parameters. Column n_J denotes the number of paths containing at least one jump in finite-activity case.

time intervals, such as 10-min or 30-min windows, thereby allowing us to pinpoint the timing of jumps. A cautious econometrician may note that the ratio tests $\hat{H}$ and $\hat{J}$ in BNS can likewise be implemented over short horizons. However, such an application raises concerns regarding the validity of the corresponding central limit theorem, since Δ_n may no longer be sufficiently small relative to the total span T . To investigate this issue, our next simulation experiment fixes the sampling frequency at one minute while progressively shrinking the total time span T . Table 4 reports the results. Specifically, BNS- $\hat{H}$ is the ratio of $(BV/RV - 1)$ to $(\widehat{IQ}/BV^2)^{1/2}$ which is the standard error to ensure $\hat{H}$ weakly converge to standard Gaussian up to some constant, and BNS- $\hat{J}$ adjust $\hat{H}$ by robustifying the standard error by setting the safeguard $\sqrt{\max(T^{-1}, \widehat{IQ}/BV^2)}$ since $\int_0^T \sigma^4 ds / (\int_0^T \sigma^2 ds)^2 \geq T^{-1}$. By Table (4), we find BNS- $\hat{H}$ has size distortion that worsens as T shrinks, and BNS- $\hat{J}$ partially, not perfectly, remedy the overrejection by touching the bound $1/T$ except when $T = 1$, which means the standard error $(\widehat{IQ}/BV^2)^{1/2}$ is a poor approximation for the standard deviation in this “finite-sample” problem and at least T^{-1} as a standard error correction is not theoretically satisfactory. Nonetheless, our fixed- k test maintains good size performance across all T ’s since our theory only requires a small Δ_n and allows testing over short blocks, which is

attributed to the coupling idea. Additionally, the fixed- k test with $k = 10$ is powerful across against finitely active jumps of rate 0.058 across all T 's. Therefore, choosing $k = 10$ is recommended for empirical practice, especially for the typical 1-min high-frequency financial data, as it generates relatively good size and power performance according to our simulations.

Table 4: Size and power as the test interval T shrinks at fixed 1-min sampling.

T	n	Size: no jumps				Power: finite-activity jumps				
		BNS- $\hat{H}$	BNS- $\hat{J}$	$k=6$	$k=10$	n_J	BNS- $\hat{H}$	BNS- $\hat{J}$	$k=6$	$k=10$
1 day	390	8.81	8.80	6.02	5.61	566	90.99	90.99	68.73	84.28
1/2 day	195	8.86	8.67	5.02	5.49	277	92.06	92.06	73.65	87.36
1/4 day	98	8.83	8.15	5.09	5.27	127	91.34	91.34	78.74	84.25
1/8 day	49	9.06	7.40	5.46	5.69	694	92.94	92.65	77.67	70.61
1/13 day	30	9.49	6.90	5.20	5.55	412	90.29	90.29	78.64	87.86
1/39 day	10	12.77	6.75	5.10	5.25	143	93.71	92.31	57.34	91.61

Note: the numbers are reported as percentage. We let $\Delta_n = 1/390$ (1-min) throughout and the time interval of interest $[0, T]$ shrink from 1 day to 10-min. The DGP follows (7) in its with-diurnal effects and finite-activity jumps specification. The computation of power is conditional on at least one jump occurring in $[0, T]$, and column n_J is the realised-jump-path count. BNS- $\hat{H}$ & $\hat{J}$ are computed over $[0, T]$: BNS- $\hat{H}$ is the ratio proposed by Eq.(11) of BNS; BNS- $\hat{J}$ is the adjusted ratio test that exploits the feasible studentiser $\max(T^{-1}, \widehat{IQ}/BV^2)$ to enforce the deterministic bound $\int_0^T \sigma^4 ds / (\int_0^T \sigma^2 ds)^2 \geq T^{-1}$, see Eq.(14) of BNS. Our fixed- k tests at $k=6, 10$ are reported. The simulation results for $T = 1, 1/2, 1/4$ are based on 10,000 trials, and for $T = 1/8, 1/13, 1/39$ are based on 100,000 trials.

7 Empirical Application: FOMC Announcement Jumps

In this section, we apply the proposed fixed- k bipower jump test to assess whether or not Federal Open Market Committee (FOMC) announcements, one of the most important macro-policies, triggered contemporaneous asset-price jumps. To do this, we collect S&P 500 E-mini futures price data at 1-minute frequency⁶ over the years 2000–2024, and the temporal data about FOMC announcement times by far are available on Bloomberg database, which provides a sample of $N = 201$ FOMC days entering the baseline analysis.⁷ Additional robustness checks using SPY

⁶For very liquid assets, such as S&P 500 E-mini futures and the ETF SPY, the 1-min frequency is sparse enough to avoid microstructure noise contamination while sufficiently frequent for asymptotic theory kicks in; see [Liu, Patton, and Sheppard \(2015\)](#). The high-frequency data are obtained from TickData: <https://www.tickdata.com>.

⁷The 201 days are 198 regularly scheduled meetings plus the 3 weekday inter-meeting actions (2001-01-03, 2001-04-18, 2020-03-03) announced within regular trading hours. Other policy actions are dropped out for they sit outside the regular trading hours.

one-minute prices are reported in Appendix C.3.

We study news-driven jumps in an event-study design framework. For each of the 201 FOMC announcement days, we first identify the exact minute at which the information is released and then consider a 30-minute window centered around the announcement, spanning from 10 minutes before to 20 minutes after the release, in line with the practice of [Nakamura and Steinsson \(2018\)](#). For the fixed- k bipower jump test, we set $k = 10$ for two reasons. First, this choice partitions the 30-minute window into three 10-minute blocks, each containing ten one-minute log-returns, and places the announcement minute at the first observation of the second block. This alignment accommodates the possibility that the announcement triggers an abrupt volatility shift at the transition from the pre-announcement block to the announcement block as shown by [Bollerslev, Li, and Xue \(2018\)](#). Second, Table 2 shows that $k = 10$ delivers relatively favorable size–power performance under this specification. We then implement the three test variants in Algorithm 1: (i) the block-level test applied to each of the three blocks within the 30-min window; (ii) the window-level max-test across the $m=3$ blocks; and (iii) the day-level test over the $m=39$ blocks spanning the trading day. For comparison, the $\hat{H}$ and $\hat{J}$ tests of BNS are occasionally reported.

To provide an instructive illustration of how financial markets assess monetary policy, we first examine four representative FOMC announcements in 2024 during the “Powell era”, before turning to the aggregate evidence. Figure 5 plots cumulative 1-minute log-returns of S&P 500 E-mini futures from the 09:30 ET opening price on these FOMC announcement days. Each panel contains the same annotation box. It reports the fixed- k bipower jump statistics for the three contiguous 10-minute blocks that tile the $[-10, +20]$ minute event window, denoted by $T_{30m,i}$ for $i = 1, 2, 3$; the Šidák window statistic T_{30m} over the 30-minute event window; the day-level fixed- k statistic T_{day} ; the day-level BNS- $\hat{J}$ statistic; and the BLL fixed- k volatility-jump statistic $\hat{c}_{1+}/\hat{c}_{1-}$.⁸

⁸Following [Bollerslev, Li, and Liao \(2021\)](#), the volatility jump test is given by

$$\frac{\hat{c}_{1+}}{\hat{c}_{1-}} \equiv \frac{\sum_{j=3}^{k'+2} (\Delta_{i_p+j}^n X)^2 \cdot 1_{\{|\Delta_{i_p+j}^n X| < u'_n\}}}{\sum_{j=3}^{k'+2} (\Delta_{i_p-j}^n X)^2 \cdot 1_{\{|\Delta_{i_p-j}^n X| < u'_n\}}}, \quad u'_n = 3\sqrt{\text{BV}_n} \Delta_n^{0.49},$$

where the i_p -th sampling interval contains the announcement p and BV_n is the realised bipower variation of that day. Under the null of no volatility uptick, the ratio $\hat{c}_{1+}/\hat{c}_{1-}$ converges in law to $F_{k',k'}$. The 5-min gap between computation

Four patterns emerge. Panel A, the September 18, 2024 50bp cut that exceeded market expectations, is the clean benchmark: the fixed- k window test T_{30m} , the day-level BNS- $\hat{J}$, and the volatility jump test $\hat{c}_{1+}/\hat{c}_{1-}$ all reject at 1%, confirming a joint price–volatility jump. What single block fixed- k bipower tests adds is timing—the price jump sits in the middle block alone as $T_{30m,2}=7.07$, significant at 1% while neither $T_{30m,1}$ nor $T_{30m,3}$ reach the 10% cutoff. Panels B and C carry the methodological point. The January 31, 2024 hawkish hold (Panel B) produces a price response confined to the announcement window: the middle block fixed- k jump test $T_{30m,2}$ rejects at 5%, while the day-level BNS- $\hat{J}$, diluted by an otherwise calm session, is essentially zero, although that price continued to decrease from 2:40 to 3pm reflected Powell pushed back expectations on policy rate cut in the conference press. The July 31, 2024 hold (Panel C) is the mirror image—elevated whole-day variability drew a BNS- $\hat{J}$ rejection with no block-level price jump with 95% confidence. The fully-priced June 12, 2024 no-move (Panel D) is the silent pole, with volatility jumping as only $\hat{c}_{1+}/\hat{c}_{1-}$ rejects. Across the four cases, the fixed- k procedure shows its advantage by tracking both the realized intraday path and the information content of the announcement: it rejects abrupt announcement-window moves, remains quiet when the release carries limited shock and price changes are muted, and helps locate any rejected jump in the relevant narrow block.

Tables 5 and 6 report the aggregate evidence. Table 5 first compares the fixed- k bipower test with the conventional BNS tests on the same FOMC event windows. At the 10-minute block level, BNS- $\hat{H}$ rejects far more often than both BNS- $\hat{J}$ and fixed- k , consistent with the size-distortion problem documented in Table 2; BNS- $\hat{J}$ is more disciplined due to truncation, but still adds a small number of rejections beyond the fixed- k rule. At the 30-minute level, however, the comparison is no longer a simple size ranking. The fixed- k statistic is a Šidák maximum over three local 10-minute tests, whereas BNS- $\hat{H}$ and BNS- $\hat{J}$ are computed once on the pooled 30-minute return series. A localized price discontinuity can therefore trigger the fixed- k window test even when the pooled BNS statistic is diluted by the remaining high-volatility part of the event window. Appendix C.2 isolates

of $\hat{c}_{1+}$ and $\hat{c}_{1-}$ is intentional to avoid news-jump contamination. We set $k' = 15$ to accommodate localness and enough observations for detecting volatility jumps. Consequently, quantiles of $F_{15,15}$ are used as critical values.

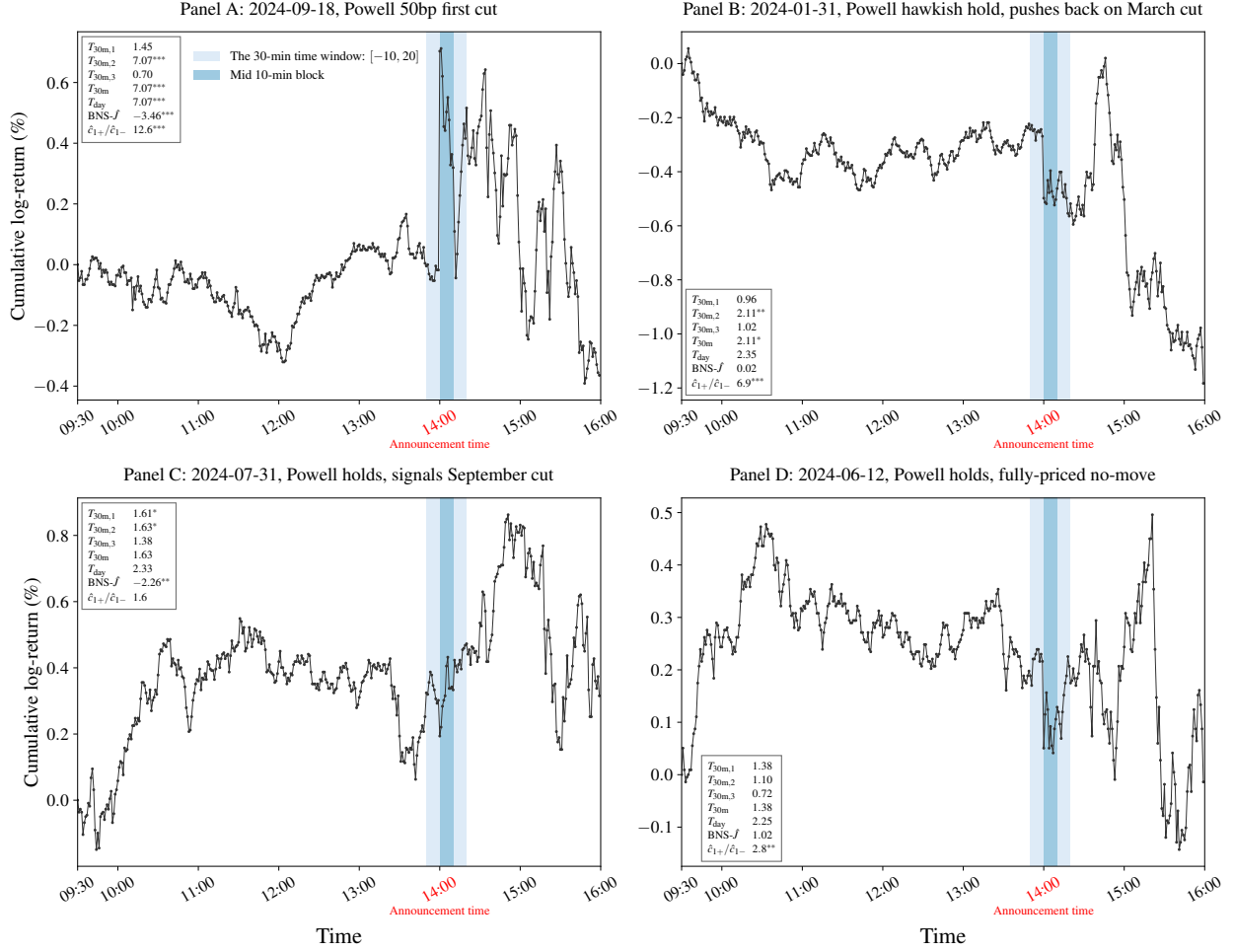


Figure 5: Cumulative 1-min log-return from the 09:30 ET open on four FOMC meetings. Each panel's annotation box reports seven statistics: (i) the fixed- k bipower price-jump test with $k=10$ at three nested levels—the three contiguous 10-min blocks tiling $[-10, +20]$ window, $T_{30m,1}$, $T_{30m,2}$, $T_{30m,3}$; their Šidák max over the 30-min window, T_{30m} ; and the Šidák max over the whole-day $m=39$ blocks, T_{day} . (ii) the day-level BNS- $\hat{J}$ adjusted ratio jump test. (iii) the fixed- k volatility jump test $\hat{c}_{1+}/\hat{c}_{1-}$. Superscripts denote significance levels: * at 10%, ** at 5%, *** at 1%.

this mechanism in a stylized 30-minute Monte Carlo experiment. This interpretation is supported by the 25 announcements for which the fixed- k window test rejects while pooled BNS- $\hat{J}$ does not: 24 of them also get rejected by the volatility-jump test $\hat{c}_{1+}/\hat{c}_{1-}$, as documented event-by-event in Table 7, indicating a post-announcement rise in volatility that can mask a localized price jump in the “pooled” BNS inference.

We therefore use the 44 fixed- k window rejections as the announcement-price-jump classification in Table 6. The cross-classification shows that price jumps and volatility jumps are related but not equivalent. Most price-jump announcements also exhibit a volatility jump: 40 of the 44 fixed- k price-jump cases fall in the price-jump/volatility-jump cell. But volatility jumps are much more common than price jumps, with 131 announcements showing volatility repricing without a fixed- k price jump. This is natural for monetary-policy news, which often changes uncertainty about the future policy path without inducing a discrete level shift in prices. The four price-jump/no-volatility-jump cases, plotted in Appendix C.3, form the opposite corner: prices move locally, but the BLL pre/post variance ratio does not reject, suggesting a level repricing that is not accompanied by a persistent post-announcement increase in volatility, or one that starts from already elevated pre-announcement variance.

Table 5: Fixed- k bipower ($k = 10$) vs. BNS- $\hat{H}$ vs BNS- $\hat{J}$ on FOMC announcements.

Testing periods	# log-returns	fixed- k rej.	BNS- $\hat{H}$ rej.	BNS- $\hat{J}$ rej.	Both fixed- k and BNS- $\hat{J}$ rej.
$T_{30m,1}$ block: $[-10, -1]$ min	10	55	98	62	54
$T_{30m,2}$ block: $[0, +9]$ min	10	25	45	28	24
$T_{30m,3}$ block: $[+10, +19]$ min	10	15	35	19	15
T_{30m} Šidák ($m=3$): $[-10, +20]$ min	30	44	39	36	19
FOMC whole days	390	48	68	68	24

Note: three jump tests on three lengths of periods are conducted on 201 weekday FOMC meetings over years 2000–2024. The table reports the rejection counts in the 201 FOMC announcements of the fixed- k bipower jump test and the BNS- $\hat{H}$ & - $\hat{J}$ tests under windows of different lengths, all at significance level $\alpha = 5\%$. For the 30-minute and whole-day rows, BNS- $\hat{H}$ and BNS- $\hat{J}$ are computed once on the pooled returns over the corresponding horizon, whereas fixed- k entries are Šidák maxima over non-overlapping 10-minute blocks. For testing jumps on FOMC whole days, BNS- $\hat{H}$ and BNS- $\hat{J}$ reject on the exact same 68 days.

Figure 6 pushes the localization idea one step further. Instead of fixing the three 10-minute blocks used for the baseline announcement-window classification, we tile the wider $[-60, +60]$

Table 6: Cross-classification of FOMC meetings: price jump $\times$ vol jump.

	Vol jump: Yes	Vol jump: No	Total
Price jump: Yes	40	4	44
Price jump: No	131	26	157
Total	171	30	201

Note: the 2-by-2 price-volatility jump classification table on $N = 201$ weekday FOMC meetings over the years 2000–2024. Price-jump rejection from the fixed- k bipower test over the $[-10, +20]$ -minute window with $k = 10$; vol-jump rejection from the $\hat{c}_{1+}/\hat{c}_{1-}$ test in [Bollerslev, Li, and Liao \(2021\)](#) on two 15-minute pre/post-announcement blocks separated by a 5-minute buffer around the announcement. Significance level is set to be 5%.

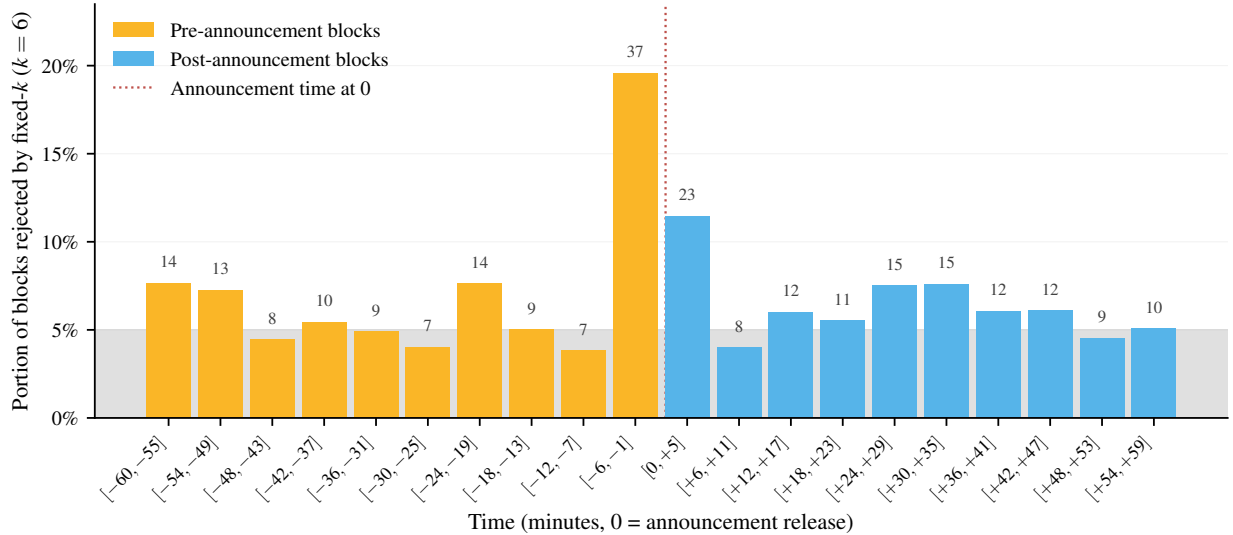


Figure 6: Fixed- k bipower jump-test rejection rates on twenty non-overlapping 6-minute blocks tiling the $[-60, +60]$ -minute window around FOMC announcements, with $k=6$ one-minute returns per block and the single-block 5% critical value $q_6^*(0.95)$. Sample: $N = 201$ weekday FOMC meetings of ES front-month 1-minute frequency price data. Each bar gives the share of meetings on which the block's $T_{n,j}$ exceeds the critical value; the integer above each bar is the rejection count on the days within the 201 meetings. The light-gray band marks the nominal 5% size; the red dotted vertical line marks the announcement instant.

minute interval around each FOMC announcement into twenty non-overlapping 6-minute blocks and apply the single-block fixed- k test with $k = 6$. This choice is deliberately conservative. A shorter block makes the local-volatility approximation sharper: volatility has less time to move within the block, the coupling error is smaller, and the resulting test is less likely to mistake smooth volatility variation for a price jump. It also gives a finer clock for locating the discontinuity once rejection occurs.

The rejection profile reveals an unexpected pattern. The largest rejection rate over the full $[-60, +60]$ -minute window occurs not after the scheduled announcement, but in the six-minute block immediately preceding it. In this block, $T_{n,j}$ rejects at the 5% level for 37 events, or 18% of the 201 meetings. This rate is more than 60% higher than the approximately 11% rejection rate observed in the immediately post-announcement block. In contrast, rejection rates in the remaining eighteen six-minute blocks stay within three percentage points of the nominal level. This concentration is consistent with the timing of monetary-policy information arrival⁹: a non-trivial share of the announcement-window price discovery has already occurred by the time the official statement is released, reflecting information leakage¹⁰, anticipatory trading, or the market processing closely related signals, with the official announcement settled to resolve the policy uncertainty and its immediate aftermath accounting for the remaining elevation in rejection frequencies. The figure thus adds the timing dimension behind Tables 5 and 6: fixed- k price-jump rejections are not scattered across the surrounding two-hour window, but cluster tightly around the announcement, with the highest mass just before the official release.

⁹The pattern of significant pre-announcement market activity is consistent with the empirical literature on pre-FOMC announcement drift. [Lucca and Moench \(2015\)](#) document a large positive drift in U.S. equity prices over the 24-hour window preceding scheduled FOMC announcements, while [Bernile, Hu, and Tang \(2016\)](#) provide evidence of informed trading in S&P 500 futures during the half-hour before scheduled macroeconomic announcements. Our fixed- k procedure complements this evidence by detecting a similar pre-announcement pattern in a much narrower six-minute pre-FOMC window.

¹⁰Within our 2000–2024 sample period, official FOMC materials themselves document concerns about the premature release of confidential monetary-policy information. At the October 23–24, 2012 meeting, Chairman Bernanke referred to an internal “leak investigation” concerning the Hilsenrath and Medley Global Advisors reports, describing the episode as “obviously damaging to the reputation and credibility of the Committee” ([Federal Open Market Committee, 2012](#), p. 3). The Committee later amended its information-security procedures “as a consequence of the 2012 leak of FOMC information” ([Federal Open Market Committee, 2016](#), p. 10). The episode was subsequently confirmed publicly when Richmond Fed President Jeffrey Lacker acknowledged having discussed confidential FOMC deliberations with Medley Global Advisors in October 2012 ([Lacker, 2017](#)).

8 Conclusion

The paper exploits a simple implication of the fixed- k coupling. Within a block containing a finite number of high-frequency “no-jump” observations, realized variance and bipower variation can be viewed as two functionals of the same local Gaussian experiment with approximately identical scale parameters. A jump shifts this coupling and cause two functionals no longer move together in the way. The ratio preserves exactly this failure while eliminating scale. Under the null of no jump, the asymptotic distribution therefore only depends on k , not on the volatility path or other nuisances. The test is simple because the reduction is complete: critical values can be computed once and tabulated, and implementation requires no quarticity estimator, truncation threshold, bandwidth choice, or bootstrap.

This fixed- k design matters because many empirical questions about jumps are local. Around an announcement, the relevant object is not the trading day as a whole, but the block in which information touches prices. Moreover, our fixed- k procedure is applicable to the whole trading day by reframing it into multiple hypothesis testing problem via the Šidák’s method. In the FOMC application, our fixed- k test helps discriminate announcement jumps from later narratives and provide alternative meaningful empirical findings. As a matter of theory, incorporating microstructure noise into the fixed- k framework is a natural next extension, so is power analysis beyond the single Brownian-scale local alternative studied here. As the jump signals may appear as densely weak which is indicated by the activity index r , rather than sparsely strong, how to combine the blockwise fixed- k jump tests and produce a uniformly powerful test is a challenging question.

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Appendix A. Proofs of Main Results

Note. In the following proof, $\|\cdot\|$ denotes the Euclidean norm, $\|\cdot\|_\infty$ denotes the sup or max norm,

Proof of Lemma 1. By the localization procedure of Section 4.4.1 in [Jacod and Protter \(2012\)](#), we may assume without loss of generality that Assumption 1 holds with $T_1 = \infty$, so that its bounds hold globally on $[0, T]$. Let $X^c \equiv X - J$ denote the continuous part of X and, for each $i \in \mathcal{I}_{n,j}$, define

$$Z_i \equiv \Delta_i^n W / \Delta_n^{1/2}, \quad \tilde{J}_i \equiv \Delta_i^n J, \quad \xi_{n,i} \equiv \Delta_i^n X^c - \sigma_{t(n,j)} \Delta_i^n W.$$

Therefore $(Z_i)_{i \in \mathcal{I}_{n,j}} \stackrel{iid}{\sim} N(0, 1)$. Using $\sigma_{t(n,j)} = \sqrt{c_{t(n,j)}}$ and we have

$$\frac{\Delta_i^n X}{\Delta_n^{1/2}} - \sqrt{c_{t(n,j)}} Z_i - \frac{\tilde{J}_i}{\Delta_n^{1/2}} = \frac{\xi_{n,i}}{\Delta_n^{1/2}}. \quad (8)$$

Specifically $\xi_{n,i} = \int_{(i-1)\Delta_n}^{i\Delta_n} b_s ds + \int_{(i-1)\Delta_n}^{i\Delta_n} (\sigma_s - \sigma_{t(n,j)}) dW_s$, by Assumption 1(i), the drift term $\int_{(i-1)\Delta_n}^{i\Delta_n} b_s ds \leq K_1 \Delta_n$; by Itô's isometry and Assumption 1(ii),

$$\mathbb{E} \left[\left(\int_{(i-1)\Delta_n}^{i\Delta_n} (\sigma_s - \sigma_{t(n,j)}) dW_s \right)^2 \right] = \mathbb{E} \left[\int_{(i-1)\Delta_n}^{i\Delta_n} |\sigma_s - \sigma_{t(n,j)}|^2 ds \right] \leq K \Delta_n^{2\kappa+1}.$$

Hence by Markov and triangle inequality, we have $|\xi_{n,i}| = O_{\mathbb{P}} \left(\Delta_n^{(\kappa+1/2) \wedge 1} \right)$ for all i . Since k is fixed, a union bound over the k indices in $\mathcal{I}_{n,j}$ yields

$$\max_{i \in \mathcal{I}_{n,j}} |\xi_{n,i} / \Delta_n^{1/2}| = O_{\mathbb{P}} \left(\Delta_n^{\kappa \wedge (1/2)} \right),$$

which via (8) establishes the claimed remainder rate. □

Proof of Theorem 1. On the null event $\Omega_{0,j}$, Lemma 1 gives $\tilde{J}_i \equiv 0$ for every $i \in \mathcal{I}_{n,j}$, so the coupling reduces to

$$\Delta_i^n X = \sqrt{c_{t(n,j)}} \Delta_n^{1/2} Z_i + R_{n,i}, \quad R_{n,i} \equiv \Delta_n^{1/2} \xi_{n,i} = O_{\mathbb{P}} \left(\Delta_n^{(\kappa+1/2) \wedge 1} \right),$$

where $(Z_1, \dots, Z_k) \stackrel{iid}{\sim} N(0, 1)$. Let $\tilde{x}_i \equiv \sqrt{c_{t(n,j)}} \Delta_n^{1/2} Z_i$, we have

$$M_n \equiv \max_{i \in \mathcal{I}_{n,j}} |\tilde{x}_i| = O_{\mathbb{P}}(\Delta_n^{1/2}) \quad R_n \equiv \max_{i \in \mathcal{I}_{n,j}} |R_{n,i}| = O_{\mathbb{P}}(\Delta_n^{(\kappa+1/2) \wedge 1}).$$

Therefore

$$\begin{aligned} \left| \sum_{i \in \mathcal{I}_{n,j}} \Delta_i^n X^2 - \sum_{i \in \mathcal{I}_{n,j}} \tilde{x}_i^2 \right| &= \left| \sum_{i \in \mathcal{I}_{n,j}} R_{n,i} (2\tilde{x}_i + R_{n,i}) \right| \leq \sum_{i=1}^k (2|\tilde{x}_i| \cdot |R_{n,i}| + R_{n,i}^2) \\ &\leq 2k M_n R_n + k R_n^2 = O_{\mathbb{P}}(\Delta_n^{(\kappa+1) \wedge 3/2}), \end{aligned}$$

while $\sum_{i \in \mathcal{I}_{n,j}} \tilde{x}_i^2 = O_{\mathbb{P}}(\Delta_n)$. Hence

$$\frac{\left| \sum_{i \in \mathcal{I}_{n,j}} \Delta_i^n X^2 - \sum_{i \in \mathcal{I}_{n,j}} \tilde{x}_i^2 \right|}{\sum_{i \in \mathcal{I}_{n,j}} \tilde{x}_i^2} = O_{\mathbb{P}}(\Delta_n^{\kappa \wedge 1/2}) = o_{\mathbb{P}}(1). \quad (9)$$

And note

$$\begin{aligned} &\left| \sum_{i=(j-1)k+2}^{jk} |\Delta_{i-1}^n X| |\Delta_i^n X| - \sum_{i=(j-1)k+2}^{jk} |\tilde{x}_{i-1}| |\tilde{x}_i| \right| \\ &= \left| \sum_{i=(j-1)k+2}^{jk} (|\Delta_{i-1}^n X| |\Delta_i^n X| - |\Delta_{i-1}^n X| |\tilde{x}_i| + |\Delta_{i-1}^n X| |\tilde{x}_i| - |\tilde{x}_{i-1}| |\tilde{x}_i|) \right| \\ &\leq \left| \sum_{i=(j-1)k+2}^{jk} (|\Delta_{i-1}^n X| |R_{n,i}| + |R_{n,i-1}| |\tilde{x}_i|) \right| \leq \sum_{i=(j-1)k+2}^{jk} (|\tilde{x}_{i-1}| |R_{n,i}| + |\tilde{x}_i| |R_{n,i-1}| + |R_{n,i-1}| |R_{n,i}|) \\ &\leq 2(k-1)M_n R_n + (k-1)R_n^2 = O_{\mathbb{P}}(\Delta_n^{(\kappa+1) \wedge 3/2}). \end{aligned} \quad (10)$$

For each block j , we define the following random variable $T_{k,j}^{\star}$ by the coupling result (3),

$$T_{k,j}^{\star} \equiv \frac{\mu_1^2(k-1)}{k} \cdot \frac{\sum_{i=(j-1)k+1}^{jk} Z_i^2}{\sum_{i=(j-1)k+2}^{jk} |Z_{i-1}| |Z_i|}$$

Therefore,

$$\sum_{i=(j-1)k+2}^{jk} |\tilde{x}_{i-1}| |\tilde{x}_i| = \frac{\mu_1^2(k-1)}{k} \cdot \frac{\sum_{i=(j-1)k+1}^{jk} \tilde{x}_i^2}{T_{k,j}^{\star}}$$

Fix $L > 0$, on the event $E_L \equiv \{T_{k,j}^\star \leq L\}$, by (10) we have

$$\begin{aligned} & \frac{\left| \sum_{i=(j-1)k+2}^{jk} |\Delta_{i-1}^n X| |\Delta_i^n X| - \sum_{i=(j-1)k+2}^{jk} |\tilde{x}_{i-1}| |\tilde{x}_i| \right|}{\sum_{i=(j-1)k+2}^{jk} |\tilde{x}_{i-1}| |\tilde{x}_i|} \\ & \leq \frac{\left| \sum_{i=(j-1)k+2}^{jk} |\Delta_{i-1}^n X| |\Delta_i^n X| - \sum_{i=(j-1)k+2}^{jk} |\tilde{x}_{i-1}| |\tilde{x}_i| \right|}{[\mu_1^2(k-1)/kL] \cdot \sum_{i=(j-1)k+1}^{jk} \tilde{x}_i^2} = O_{\mathbb{P}}(\Delta_n^{\kappa \wedge 1/2}) = o_{\mathbb{P}}(1). \end{aligned} \quad (11)$$

Combining (9) and (11) yields

$$\begin{aligned} T_{n,j} &= \frac{\mu_1^2(k-1)}{k} \cdot \frac{(\sum_{i=(j-1)k+1}^{jk} \tilde{x}_i^2)(1 + o_{\mathbb{P}}(1))}{(\sum_{i=(j-1)k+2}^{jk} |\tilde{x}_{i-1}| |\tilde{x}_i|)(1 + o_{\mathbb{P}}(1))} \\ &= \frac{\mu_1^2(k-1)}{k} \cdot \frac{\sum_{i=(j-1)k+1}^{jk} Z_i^2}{\sum_{i=(j-1)k+2}^{jk} |Z_{i-1}| |Z_i|} \cdot (1 + o_{\mathbb{P}}(1)) \\ &= T_{k,j}^\star (1 + o_{\mathbb{P}}(1)) \quad \text{on } E_L. \end{aligned} \quad (12)$$

For any $L > 0$, we have

$$E_L \supseteq \left\{ \sum_{i=(j-1)k+1}^{jk} Z_i^2 \leq c_k^{-1} L^{1/2} \right\} \cap \left\{ \sum_{i=(j-1)k+2}^{jk} |Z_{i-1}| |Z_i| > L^{-1/2} \right\}$$

implying that $E_L^c \subseteq \left\{ \sum_{i=(j-1)k+1}^{jk} Z_i^2 > c_k^{-1} \sqrt{L} \right\} \cup \left\{ \sum_{i=(j-1)k+2}^{jk} |Z_{i-1}| |Z_i| \leq L^{-1/2} \right\}$ with $c_k \equiv \mu_1^2(k-1)/k < 1$. Thus

$$\begin{aligned} \mathbb{P}(E_L^c) &\leq \mathbb{P}\left(\sum_{i=(j-1)k+1}^{jk} Z_i^2 > c_k^{-1} L^{1/2}\right) + \mathbb{P}\left(\sum_{i=(j-1)k+2}^{jk} |Z_{i-1}| |Z_i| \leq L^{-1/2}\right) \\ &\leq \mathbb{P}\left(\sum_{i=(j-1)k+1}^{jk} Z_i^2 > L^{1/2}\right) + \mathbb{P}(|Z_1| |Z_2| \leq L^{-1/2}) \\ &\stackrel{(*)}{=} O\left(L^{k/4-1/2} \exp(-L^{1/2}/2)\right) + O(L^{-1/2} \log L) \rightarrow 0, \quad \text{as } L \rightarrow \infty. \end{aligned}$$

where the first term of the RHS of equality (*) is by the tail probability of χ_k^2 and the second part is due to the density of the product of two independent folded standard normals is $f(z) = \frac{2}{\pi} K_0(z)$ where $K_0(\cdot)$ is the modified Bessel function of the second kind of order zero, see [Craig \(1936\)](#),

and $K_0(z) \approx \log(1/z)$ when $z \downarrow 0$, see 10.30.3 of [Olver, Lozier, Boisvert, and Clark \(2010\)](#), so $\int_0^{L^{-1/2}} f(z) dz \approx \frac{2}{\pi} \int_0^{L^{-1/2}} \log(1/z) dz = \frac{2}{\pi} L^{-1/2} + \frac{1}{\pi} L^{-1/2} \log(L) = O(L^{-1/2} \log L)$ when L large. Thus for all $\epsilon > 0$,

$$\mathbb{P}(|T_{n,j} - T_{k,j}^\star| > \epsilon) \leq \mathbb{P}(|T_{n,j} - T_{k,j}^\star| > \epsilon, E_L) + \mathbb{P}(E_L^c) = o(1) + \mathbb{P}(E_L^c)$$

Letting $n \rightarrow \infty$ and then $L \rightarrow \infty$ yields $T_{n,j} \xrightarrow{\mathbb{P}} T_{k,j}^\star$. Finally, because $T_{k,j}^\star$ has the same distribution as $T_k^\star$ for all j , we have $T_{n,j} \xrightarrow{d} T_k^\star$ as claimed. $\square$

Proof of Proposition 2. Set $|y| = (|y_1|, \dots, |y_k|)'$ so that $\sum_{i=1}^k y_i^2 = |y'| |y|$ and a matrix A with $A_{i,j} = \frac{1}{2} \cdot 1_{\{|i-j|=1\}}$. Therefore by the Rayleigh quotient theorem, see Theorem 4.2.2 of [Horn and Johnson \(2012\)](#), we have

$$\frac{\sum_{i=2}^k |y_{i-1}| |y_i|}{\sum_{i=1}^k y_i^2} = \frac{|y'| A |y|}{|y'| |y|} \leq \lambda_{\max}(A) = \cos(\pi/(k+1))$$

where $\lambda_{\max}(A)$ denotes A 's largest eigenvalue, and the eigenvector corresponding to $\lambda_{\max}(A)$ is $v_{\max} = (\sin \frac{\pi}{k+1}, \sin \frac{2\pi}{k+1}, \dots, \sin \frac{k\pi}{k+1})'$. The above inequality becomes tight *iff* y is a scalar multiple of $v_{\max}$, that is, $|y_i| = c \sin(i\pi/(k+1))$ for all $i = 1, \dots, k$ and some $c > 0$, so we have the desired lower bound (24) as claimed. $\square$

Proof of Proposition 1. Write $X^c \equiv X - J$ for the continuous part, so on the block $\Delta_i^n X = \Delta_i^n X^c + J \cdot 1_{\{i=i^\star\}}$. Lemma 1 gives $\Delta_i^n X^c = \sqrt{c_{t(n,j)} \Delta_n} Z_i + O_{\mathbb{P}}(\Delta_n)$ uniformly in $i \in \mathcal{I}_{n,j}$, with $(Z_i) \stackrel{iid}{\sim} N(0, 1)$. Hence

$$\sum_{i \in \mathcal{I}_{n,j}} (\Delta_i^n X)^2 = J^2 + 2J \Delta_{i^\star}^n X^c + \sum_{i \in \mathcal{I}_{n,j}} (\Delta_i^n X^c)^2 = J^2 + O_{\mathbb{P}}(\Delta_n^{1/2}),$$

so that, since $J \neq 0$ a.s.,

$$\hat{c}_{n,j}^{\text{RV}} = \frac{J^2}{k \Delta_n} (1 + O_{\mathbb{P}}(\Delta_n^{1/2})) = O_{\mathbb{P}}(\Delta_n^{-1}). \quad (13)$$

For the denominator, we have $|\Delta_{i^\star}^n X| = |J + \sqrt{c_{t(n,j)} \Delta_n} Z_{i^\star} + O_{\mathbb{P}}(\Delta_n)| = |J| (1 + O_{\mathbb{P}}(\Delta_n^{1/2}))$ by

Lemma 1. Since $i^\star$ is interior, J enters exactly two cross terms. Hence

$$|\Delta_{i^\star-1}^n X| |\Delta_{i^\star}^n X| + |\Delta_{i^\star}^n X| |\Delta_{i^\star+1}^n X| = |J| \sqrt{c_{t(n,j)} \Delta_n} (|Z_{i^\star-1}| + |Z_{i^\star+1}|) (1 + o_{\mathbb{P}}(1)),$$

while each of the remaining $k - 3$ bipower products is $c_{t(n,j)} \Delta_n |Z_{i-1}| |Z_i| (1 + o_{\mathbb{P}}(1)) = O_{\mathbb{P}}(\Delta_n)$.

Therefore

$$\hat{c}_{n,j}^{\text{BV}} = \frac{\mu_1^{-2}}{(k-1)\sqrt{\Delta_n}} |J| \sqrt{c_{t(n,j)}} (|Z_{i^\star-1}| + |Z_{i^\star+1}|) (1 + o_{\mathbb{P}}(1)) = O_{\mathbb{P}}(\Delta_n^{-1/2}). \quad (14)$$

Taking the ratio (13) to (14) yields the displayed leading order expression for $T_{n,j}$. The distributional equivalence follows because the Lemma 1 shocks (Z_i) are iid $N(0, 1)$ independent of $(c_{t(n,j)}, J)$. $\square$

Proof of Theorem 2. By Lemma 1, for each $j \in \mathcal{M}$ there exists a Gaussian vector $Z^{(j)} \equiv (Z_i)_{i \in \mathcal{I}_{n,j}}$ with $Z_i \equiv \Delta_i^n W / \Delta_n^{1/2} \stackrel{\text{iid}}{\sim} N(0, 1)$ for $i \in \mathcal{I}_{n,j}$ such that, on the joint null $\cap_{j \in \mathcal{M}} \Omega_{0,j}$,

$$\max_{i \in \mathcal{I}_{n,j}} \left| \Delta_i^n X / \Delta_n^{1/2} - \sqrt{c_{t(n,j)}} Z_i \right| = O_{\mathbb{P}}(\Delta_n^{1/2}), \quad j \in \mathcal{M}.$$

Because Brownian increments on disjoint intervals are independent, the vectors $(Z^{(j)})_{j \in \mathcal{M}}$ are mutually independent across $j \in \mathcal{M}$. For $z = (z_1, \dots, z_k)'$, define $\mathcal{R}(z) \equiv \frac{\mu_1^2(k-1)}{k} \cdot \sum_{i=1}^k z_i^2 / \sum_{i=2}^k |z_{i-1}| |z_i|$.

By the proof of Theorem 1 and independence, we have the joint convergence in probability

$$(T_{n,j})_{j \in \mathcal{M}} \xrightarrow{\mathbb{P}} (\mathcal{R}(Z^{(j)}))_{j \in \mathcal{M}} \stackrel{d}{=} (T_k^{\star,(j)})_{j \in \mathcal{M}}, \quad \text{where } T_k^{\star,(j)} \stackrel{\text{iid}}{\sim} T_k^{\star}.$$

The product distribution function and Šidák correction follow by independence in the limit,

$$\mathbb{P}(T_n^{\text{max}, \mathcal{M}} \leq x) \rightarrow \prod_{j \in \mathcal{M}} \mathbb{P}(T_k^{\star,(j)} \leq x) = F_{T_k^{\star}}(x)^m.$$

Setting $F_{T_k^{\star}}(x)^m = 1 - \alpha$ inverts to $x = q_k^{\star}((1 - \alpha)^{1/m}) \equiv \text{cv}_{\alpha,k,m}$. $\square$

Proof of Theorem 3. Write

$$\kappa_0 \equiv \kappa \wedge \frac{1}{2}, \quad c_k \equiv \frac{\mu_1^2(k-1)}{k}, \quad d_k \equiv \lfloor k/2 \rfloor.$$

The displayed block-size condition $k \geq 2 \left\lfloor \frac{1}{\kappa \wedge (1/2)} \right\rfloor + 2$ is exactly $1/d_k < \kappa_0$. The proof has three steps. The first obtains a uniform scale-free Gaussian coupling. The second records the tail geometry of the Gaussian rejection cone. The third controls the probability that the coupling error moves any block across the boundary of that cone. Throughout the proof, we extend $\mathcal{R}(z)$ to equal $+\infty$ when its bipower denominator vanishes and $z \neq 0$; its value at $z = 0$ is immaterial. These conventions concern Gaussian null sets and make the rejection-cone notation unambiguous.

Step 1. Uniform scale-free coupling. By the standard stopped-process localization, it is enough to prove the result under deterministic bounds

$$|b_t| \leq B, \quad B^{-1} \leq |\sigma_t| \leq B, \quad t \in [0, 1],$$

for every fixed B , and then let $B \rightarrow \infty$. Set $t_j \equiv t(n, j)$. On the joint null, for $i \in \mathcal{I}_{n,j}$ write

$$\xi_{n,i}^{(j)} \equiv \frac{\Delta_i^n X}{\Delta_n^{1/2}} - \sigma_{t_j} Z_i^{(j)}, \quad Z_i^{(j)} \equiv \frac{\Delta_i^n W}{\Delta_n^{1/2}}.$$

The vectors $Z^{(j)} \equiv (Z_i^{(j)})_{i \in \mathcal{I}_{n,j}}$ are mutually independent and each is $N(0, I_k)$. For every integer $p \geq 2$, the Burkholder–Davis–Gundy and Hölder inequalities and Assumption 2 give, uniformly in (n, i, j) ,

$$\begin{aligned} \mathbb{E} |\xi_{n,i}^{(j)}|^p &\leq C_{B,p} \Delta_n^{p/2} + C_p \Delta_n^{-p/2} \mathbb{E} \left[\left(\int_{(i-1)\Delta_n}^{i\Delta_n} |\sigma_s - \sigma_{t_j}|^2 ds \right)^{p/2} \right] \\ &\leq C_{B,p} \Delta_n^{p/2} + C_p \Delta_n^{-1} \int_{(i-1)\Delta_n}^{i\Delta_n} \mathbb{E} |\sigma_s - \sigma_{t_j}|^p ds \leq C_{B,p,k} \Delta_n^{p\kappa_0}. \end{aligned}$$

Define the normalized block error

$$e_{n,j} \equiv \left(\frac{\xi_{n,i}^{(j)}}{\sigma_{t_j}} \right)_{i \in \mathcal{I}_{n,j}}.$$

Because $\mathcal{R}(az) = \mathcal{R}(z)$ for every $a \neq 0$, the statistic has the exact representation

$$T_{n,j} = \mathcal{R} \left(\left(\frac{\Delta_i^n X}{\sigma_{t_j} \Delta_n^{1/2}} \right)_{i \in \mathcal{I}_{n,j}} \right) = \mathcal{R}(Z^{(j)} + e_{n,j}). \quad (15)$$

Choose a constant ζ such that

$$\frac{1}{d_k} < \zeta < \kappa_0$$

and then an integer $p \geq 2$ such that $p(\kappa_0 - \zeta) > 1$. Since $m_n \asymp \Delta_n^{-1}$, Markov's inequality and a union bound yield, for $\varepsilon_n \equiv m_n^{-\zeta}$,

$$\mathbb{P}\left(\max_{j \leq m_n} \|e_{n,j}\| > \varepsilon_n\right) \leq C_{B,p,k} m_n \Delta_n^{p\kappa_0} \varepsilon_n^{-p} \leq C_{B,p,k} m_n^{1-p(\kappa_0-\zeta)} \longrightarrow 0. \quad (16)$$

Step 2. Tail geometry of the Gaussian rejection cone. For $z \neq 0$, set

$$N(z) \equiv \sum_{i=1}^k z_i^2, \quad D(z) \equiv \sum_{i=2}^k |z_{i-1}| |z_i|, \quad S(z) \equiv \frac{D(z)}{N(z)} = \frac{c_k}{\mathcal{R}(z)}.$$

For $a > 0$, define the closed cone

$$\mathcal{K}_k(a) \equiv \{0\} \cup \{z \in \mathbb{R}^k \setminus \{0\} : S(z) \leq a\}, \quad G_k(a) \equiv \mathbb{P}(Z \in \mathcal{K}_k(a)),$$

where $Z \sim N(0, I_k)$. We use the following fixed- k geometric fact:

$$G_k(a) = C_k a^{d_k} \{\log(1/a)\}^{s_k} \{1 + o(1)\}, \quad a \downarrow 0, \quad (17)$$

for some $C_k \in (0, \infty)$ and finite integer $s_k \geq 0$. Moreover, G_k is continuous and, for any deterministic sequences $a_n \downarrow 0$ and $h_n/a_n \rightarrow 0$,

$$\mathbb{P}(Z \in \mathcal{K}_k(a_n)^{h_n} \setminus \mathcal{K}_k(a_n)) = o(G_k(a_n)), \quad (18)$$

where $A^h \equiv \{z : \text{dist}(z, A) \leq h\}$.

For completeness, we verify this fact. Write $Z = R_Z U$, where $R_Z = \|Z\|$ is independent of $U = Z/\|Z\|$, and U is uniform on the unit sphere. On that sphere,

$$S(U) = H_k(U) \equiv \sum_{i=2}^k |U_{i-1}| |U_i|.$$

The random variable $H_k(U)$ has no atoms. Indeed, after splitting the sphere into its finitely many

sign orthants, every positive level set is the zero set of a nonconstant real-analytic function and hence has sphere measure zero; the zero level is a finite union of coordinate subspheres indexed by the independent sets of the path graph on k vertices.

We next identify its lower-tail exponent. Let $\bar{S} \equiv \sum_{i=1}^k Z_i^2$ and $D_Z \equiv \sum_{i=2}^k |Z_{i-1}||Z_i|$. For $M_a = A \log(1/a)$ and $A > 2d_k$,

$$\begin{aligned} G_k(a) &\leq \mathbb{P}(\bar{S} > M_a) + \mathbb{P}(D_Z \leq aM_a) \\ &\leq \mathbb{P}(\bar{S} > M_a) + \prod_{\ell=1}^{d_k} \mathbb{P}(|Z_{2\ell-1}||Z_{2\ell}| \leq aM_a) \leq \bar{C}_k a^{d_k} \{\log(1/a)\}^{2d_k}. \end{aligned}$$

The last inequality uses the chi-squared upper-tail bound and $\mathbb{P}(|Z_1||Z_2| \leq x) \leq Cx \log(1/x)$ for small x ; the latter follows from the product density $(2/\pi)K_0(x)$ ([Craig, 1936](#)) and the expansion of K_0 at zero ([Olver, Lozier, Boisvert, and Clark, 2010](#), 10.30.3). For the reverse bound, choose a maximum independent set I of the path graph and put $V = I^c$, so $|V| = d_k$. Let w range over a compact interior patch of the coordinate sphere on I , let $v \in [0, da]^{|V|}$, and set

$$U_I = \sqrt{1 - \|v\|^2} w, \quad U_V = v,$$

where $d > 0$ is a sufficiently small fixed constant. The surface Jacobian of this parametrization is bounded below uniformly. Every edge has an endpoint in V , so $H_k(U) \leq a$, while this set has sphere measure bounded below by $\underline{C}_k a^{d_k}$. Consequently,

$$\underline{C}_k a^{d_k} \leq G_k(a) \leq \bar{C}_k a^{d_k} \{\log(1/a)\}^{2d_k} \quad \text{for all sufficiently small } a. \quad (19)$$

To pass from the two-sided order bound to (17), restrict first to the positive orthant and take a finite smooth partition of unity subordinate to projective charts on which one coordinate is bounded below by a fixed multiple of the largest coordinate. In chart j , set $x_j = 1$ and $x_i = |u_i|/|u_j|$ for $i \neq j$. Each localized contribution is then a real-analytic sublevel integral against a smooth compactly

supported weight on a bounded manifold-with-corners chart, and

$$H_k(u) = \frac{\sum_{i=2}^k x_{i-1}x_i}{\sum_{i=1}^k x_i^2}.$$

After a further finite localization around the compact zero set in each chart, the standard real-analytic sublevel-volume expansion applies term by term and gives a leading term $Ca^\lambda \{\log(1/a)\}^s$ for some $\lambda > 0$ and finite integer $s \geq 0$.¹¹ Summing the finitely many localized contributions preserves this form. The bounds in (19) force $\lambda = d_k$, which proves (17).

It remains to prove (18). The map H_k is 2-Lipschitz on the sphere: expanding each adjacent product and applying the Cauchy–Schwarz inequality twice gives

$$|H_k(u) - H_k(v)| \leq 2\|u - v\|, \quad u, v \in \mathbb{S}^{k-1}.$$

Suppose $z \in \mathcal{K}_k(a)^h \setminus \mathcal{K}_k(a)$, $\rho \equiv \|z\| > 2h$, and choose $y \in \mathcal{K}_k(a)$ with $\|z - y\| \leq h$. Then

$$\left\| \frac{z}{\rho} - \frac{y}{\|y\|} \right\| \leq \frac{2h}{\rho}, \quad H_k\left(\frac{z}{\rho}\right) \leq a + \frac{4h}{\rho}.$$

It follows from the polar decomposition that

$$\mathbb{P}(Z \in \mathcal{K}_k(a)^h \setminus \mathcal{K}_k(a)) \leq \mathbb{P}(R_Z \leq 2h) \tag{20}$$

$$+ \mathbb{E} \left[\left\{ G_k\left(a + \frac{4h}{R_Z}\right) - G_k(a) \right\} \mathbf{1}_{\{R_Z > 2h\}} \right]. \tag{21}$$

Set $\omega = h/a$. By (17), G_k is regularly varying at zero with index d_k . Hence, for every fixed $R_Z > 0$, the expectation's integrand divided by $G_k(a)$ tends to zero when $\omega \rightarrow 0$. Choose $\nu \in (d_k, k)$. We use the following global version of the standard bound for a regularly varying function (see [Haan and Ferreira 2006](#), Chapter 1): uniformly over $x \geq 1$ and all sufficiently small a ,

$$\frac{G_k(ax)}{G_k(a)} \leq C_\nu x^\nu.$$

To see that the bound holds for all $x \geq 1$, fix a sufficiently small a_0 . If $ax \leq a_0$, it is the usual local

¹¹For the real-analytic sublevel expansion used here, see Michael Greenblatt, “Resolution of singularities, asymptotic expansions of integrals and related phenomena,” *Journal d'Analyse Mathématique* 111 (2010), 221–245.

bound. If $ax > a_0$, then $G_k(ax) \leq 1$, while (17), $x > a_0/a$, and $\nu > d_k$ imply $G_k(a)^{-1} \leq C_\nu x^\nu$. Therefore the normalized integrand in (21) is dominated by $C_\nu(1 + R_Z^{-\nu})$, which is integrable because $\mathbb{E}[R_Z^{-\nu}] < \infty$ for $\nu < k$. Dominated convergence shows that the expectation is $o(G_k(a))$. Finally, $\mathbb{P}(R_Z \leq 2h) = O(h^k) = o(G_k(a))$ whenever $h/a \rightarrow 0$, since $k > d_k$ and the remaining factor in (17) is slowly varying. This proves (18).

Step 3. Critical cone and rejection probability. Put

$$p_n \equiv (1 - \alpha)^{1/m_n}, \quad q_n \equiv q_k^\star(p_n), \quad a_n \equiv \frac{c_k}{q_n}.$$

The lower bound in (19) shows that $T_k^\star$ has unbounded support. Hence $p_n \uparrow 1$ implies $q_n \rightarrow \infty$ and $a_n \downarrow 0$. Moreover, the absence of atoms established in Step 2 makes the cdf of $T_k^\star$ continuous, so the generalized-inverse convention for $q_k^\star$ gives $F_{T_k^\star}(q_n) = p_n$. Consequently,

$$G_k(a_n) = \mathbb{P}(T_k^\star > q_n) = 1 - p_n \sim \frac{-\log(1 - \alpha)}{m_n}. \quad (22)$$

By (17) and inversion of a regularly varying function,

$$a_n = m_n^{-1/d_k + o(1)}.$$

The choices in Step 1 therefore give

$$\frac{\varepsilon_n}{a_n} = m_n^{-(\zeta - 1/d_k) + o(1)} \rightarrow 0. \quad (23)$$

Let $E_n \equiv \{\max_{j \leq m_n} \|e_{n,j}\| \leq \varepsilon_n\}$. On E_n , the following inclusion is deterministic:

$$\begin{aligned} \left\{ \max_{j \leq m_n} \mathcal{R}(Z^{(j)} + e_{n,j}) > q_n \right\} &\subseteq \left\{ \exists j \leq m_n : Z^{(j)} \in \mathcal{K}_k(a_n) \right\} \\ &\cup \bigcup_{j=1}^{m_n} \left\{ Z^{(j)} \in \mathcal{K}_k(a_n)^{\varepsilon_n} \setminus \mathcal{K}_k(a_n) \right\}. \end{aligned}$$

Indeed, if $Z^{(j)} + e_{n,j}$ belongs to the rejection cone while $Z^{(j)}$ does not, then $Z^{(j)}$ is within ε_n of that

cone. The first event on the right has probability exactly

$$1 - \{1 - G_k(a_n)\}^{m_n} = 1 - p_n^{m_n} = \alpha$$

by independence of the Gaussian blocks and (22). By (18), (23), and a union bound, the probability of the second event is

$$m_n o(G_k(a_n)) = o(1).$$

Combining this result with (15) and (16) gives

$$\mathbb{P}(T_n^{\max} > q_n) \leq \alpha + o(1)$$

for every fixed localization level B . Letting $B \rightarrow \infty$ proves the claim. $\square$

Proof of Theorem 4. Write $t^\star \equiv t(n, j^\star)$ and $\sigma^\star \equiv \sigma_{t^\star}$. By localization we may restrict attention to events on which $\sigma^\star$ is bounded and bounded away from zero; the localization error can be sent to zero at the end. Since $\eta_n \in \mathcal{T}_{n, j^\star}$ and k is fixed, Assumption 1(ii) implies $\sigma_{\eta_n}/\sigma^\star \rightarrow 1$ in probability. Under $H_{1,n}^{\text{ann}}(\gamma)$, the jump contribution appears only in return $r_n^\star = (j^\star - 1)k + L_n$. Lemma 1 therefore gives, uniformly over within-block coordinates $\ell = 1, \dots, k$,

$$\frac{\Delta_{(j^\star-1)k+\ell}^n X}{\sigma^\star \Delta_n^{1/2}} = Z_\ell + \gamma \mathbf{1}_{\{\ell=L_n\}} + o_{\mathbb{P}}(1),$$

where $Z = (Z_1, \dots, Z_k)$ has iid $N(0, 1)$ components.

The statistic $T_{n, j^\star}$ is exactly $\mathcal{R}$ applied to the vector of block returns, and $\mathcal{R}$ is scale invariant: $\mathcal{R}(ay) = \mathcal{R}(y)$ for every $a > 0$. Hence the factor $\sigma^\star \Delta_n^{1/2}$ cancels. Moreover, $\mathcal{R}$ is continuous at $Z + \gamma e_L$ with probability one, since the denominator

$$\sum_{i=2}^k |Z_{i-1} + \gamma \mathbf{1}_{\{i-1=L\}}| |Z_i + \gamma \mathbf{1}_{\{i=L\}}|$$

is zero only on a finite union of Gaussian hyperplanes. The deterministic/independent timing

condition gives the joint convergence $(Z, L_n) \xrightarrow{d} (Z, L)$, so the continuous mapping theorem yields

$$T_{n,j^\star} \xrightarrow{d} G_{\gamma,L}.$$

This proves the distributional statement in part (i). Since $G_{\gamma,L}$ has no atom at $q_k^\star(1 - \alpha)$, the portmanteau theorem gives

$$\lim_{n \rightarrow \infty} \mathbb{P}_{H_{1,n}^{\text{ann}}(\gamma)}(T_{n,j^\star} > q_k^\star(1 - \alpha)) = \mathbb{P}(G_{\gamma,L} > q_k^\star(1 - \alpha)),$$

which is the weighted sum in the theorem after conditioning on L .

For part (ii), apply the same coupling block by block on the fixed collection $\mathcal{M}_n$. The contaminated block converges to $G_{\gamma,L}$, while each continuous block converges to an independent copy of $T_k^\star$. Hence

$$T_n^{\max, \mathcal{M}_n} \xrightarrow{d} \max\{G_{\gamma,L}, T_k^{\star,(1)}, \dots, T_k^{\star,(m-1)}\}.$$

Evaluating the probability that this maximum is no larger than $c_{k,m}(\alpha)$ gives

$$\mathbb{P}(G_{\gamma,L} \leq c_{k,m}(\alpha)) F_k(c_{k,m}(\alpha))^{m-1},$$

and conditioning on L gives the displayed power formula.

For part (iii), $c_{k,m_n}(\alpha) \rightarrow \infty$ because $(1 - \alpha)^{1/m_n} \rightarrow 1$. Since $G_{\gamma,L}$ is finite almost surely,

$$\mathbb{P}(G_{\gamma,L} > c_{k,m_n}(\alpha)) \rightarrow 0.$$

Let $T_{n,-j^\star}^{\max} \equiv \max_{j \leq m_n, j \neq j^\star} T_{n,j}$. By the union bound,

$$\mathbb{P}_{H_{1,n}^{\text{ann}}(\gamma)}(T_n^{\max} > c_{k,m_n}(\alpha)) \leq \mathbb{P}(T_{n,-j^\star}^{\max} > c_{k,m_n}(\alpha)) + \mathbb{P}(T_{n,j^\star} > c_{k,m_n}(\alpha)).$$

The second probability tends to zero by part (i) and $c_{k,m_n}(\alpha) \rightarrow \infty$. The first is bounded asymptotically by α by Theorem 3, applied to the continuous blocks and the conservative threshold

based on m_n rather than $m_n - 1$. In the iid Gaussian reference experiment the rejection probability is

$$1 - \mathbb{P}(G_{\gamma,L} \leq c_{k,m_n}(\alpha)) F_k(c_{k,m_n}(\alpha))^{m_n-1},$$

which tends to $1 - (1 - \alpha) = \alpha$. □

Appendix B. Additional Propositions

B.1 A Deterministic Lower Bound

Proposition 2 (Sharp lower bound). *For real numbers $\{y_i\}_{i=1}^k$ with $\sum_{i=2}^k |y_{i-1}| |y_i| > 0$,*

$$\frac{\mu_1^2(k-1)}{k} \cdot \frac{\sum_{i=1}^k y_i^2}{\sum_{i=2}^k |y_{i-1}| |y_i|} \geq \frac{\mu_1^2(k-1)}{k \cos(\pi/(k+1))} \equiv L_k, \quad (24)$$

with equality if and only if $|y_i| = c \sin(i\pi/(k+1))$ for all $i = 1, \dots, k$ and some constant $c > 0$. Consequently, both the test statistic $T_{n,j}$ and its null limit $T_k^\star$ are almost surely no smaller than the positive constant L_k .

This bound has two consequences that drive everything below. (i) *Finite-sample lower bound on the statistic.* For every path and every block, $T_{n,j}$ lives in the half-line $[\mu_1^2(k-1)/[k \cos(\pi/(k+1))], \infty)$. For $k = 6$ this endpoint is 0.589; for $k = 10$ it is 0.597; for $k = 15$ it is 0.606. The asymptotic null distribution $T_k^\star$ is concentrated slightly above this endpoint, and the jump alternative pushes the statistic arbitrarily far to the right. In effect, $T_{n,j}$ is naturally a one-sided statistic whose lower tail is *pinned* by arithmetic, not by randomness. (ii) *Regularity of the functional on the Gaussian block.* The map $(z_1, \dots, z_k) \mapsto \mathcal{R}(z) \equiv \mu_1^2(k-1)/k \cdot \sum z_i^2 / \sum |z_{i-1}| |z_i|$ has singularities only on the Lebesgue null set $\{\sum |z_{i-1}| |z_i| = 0\}$. Away from that set, T is continuous. Crucially, because the *ratio* is bounded below, no auxiliary density-at-zero control on the denominator is needed; the remaining uniform argument only requires controlling the polynomial upper tail of $T_k^\star$.

Appendix C. Additional Simulations and Empirical Results

C.1. A local-power scale diagnostic

Figure 7 illustrates the rate distinction behind Theorem 4. Each Monte Carlo draw contains n Gaussian returns

$$R_i = \Delta_n^{1/2} Z_i + J_n 1_{\{i=\lfloor n/2 \rfloor\}}, \quad Z_i \stackrel{iid}{\sim} N(0, 1), \quad \Delta_n = 1/n,$$

with $n \in \{250, 500, 1000, 2000, 4000\}$. The fixed- k tests use $k = 10$ and appear in three versions that mirror parts (i)–(iii) of Theorem 4: the block test applied to the block containing the simulated jump, with the 5% upper critical value of T_{10}^* ; the Šidák window test over $m = 3$ blocks containing that block, with critical value $q_{10}^*(0.95^{1/3})$; and the day-level scan over all $m_n = n/10$ blocks, with critical value $q_{10}^*(0.95^{1/m_n})$. The jump lands in the interior of its block. BNS- $\hat{G}$ and BNS- $\hat{J}$ are computed over the full n -return window and use Monte Carlo size-adjusted lower-tail critical values under the no-jump null, so the comparison is not driven by size distortion. Because the returns are exactly Gaussian in this design, the block and window rejection rates are n -free by construction and coincide with the limiting power of Theorem 4; the day-level scan, whose critical value grows with m_n , is the only fixed- k line with genuine n -dependence.

The left panel sets $J_n = 4\Delta_n^{1/2}$. This is the Brownian scale of the fixed- k experiment. The block test holds at 25.6% and the window test at 15.2%, flat in n , while the day-level scan decays from 7.1% at $n = 250$ toward the 5% size as its Šidák critical value diverges: the three lines realize parts (i)–(iii) of Theorem 4. For BNS- $\hat{G}$ and BNS- $\hat{J}$, the jump enters the large-window statistic through the normalised discrepancy $\Delta_n^{-1/2}(RV - BV)$, and its contribution is only of order $\Delta_n^{-1/2}J_n^2 = O(\Delta_n^{1/2})$; their rejection probabilities therefore return toward size. The right panel sets $J_n = 0.75\Delta_n^{1/4}$, the BNS Pitman scale. The BNS tests then have nontrivial power. The fixed- k block test also gains power, but for a different reason: relative to the block Brownian scale $\Delta_n^{1/2}$, the jump displacement is $0.75\Delta_n^{-1/4}$ and therefore grows with n . The figure is thus a rate diagnostic, not a claim that one

statistic uniformly dominates the other.

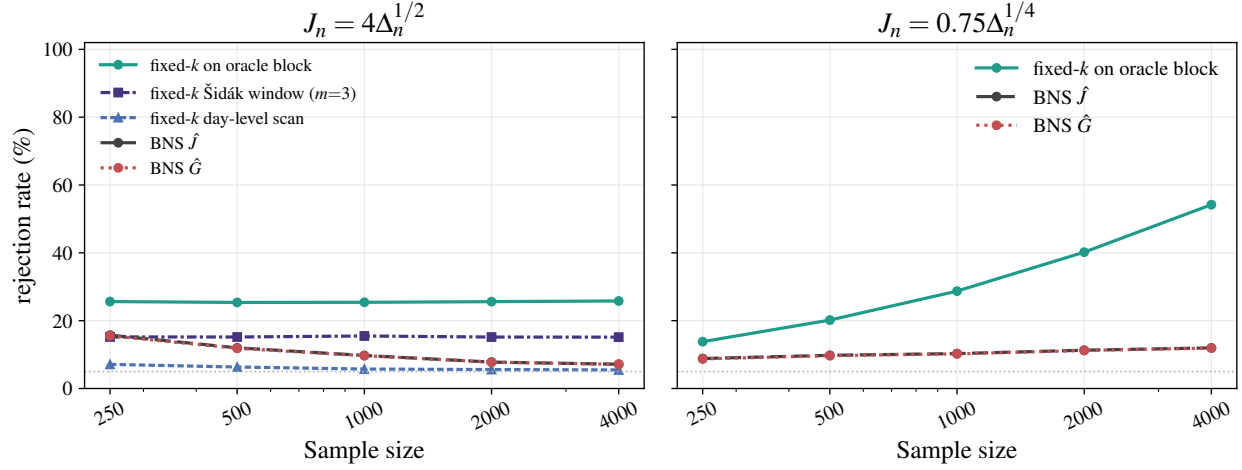


Figure 7: A local-power scale diagnostic. The left panel uses the Brownian-scale alternative $J_n = 4\Delta_n^{1/2}$; the right panel uses the BNS-scale alternative $J_n = 0.75\Delta_n^{1/4}$. The fixed- k lines use $k = 10$: the block test on the block containing the simulated jump; in the left panel additionally the Šidák window test over $m = 3$ blocks and the day-level scan over all $m_n = n/10$ blocks, realizing parts (i)–(iii) of Theorem 4. The jump lands in the interior of its block. BNS- $\hat{G}$ and BNS- $\hat{J}$ are computed on the full n -return window with Monte Carlo size-adjusted 5% critical values. Each alternative point uses 50,000 replications; the BNS null critical values use 70,000 replications and the fixed- k Šidák critical values 8,000,000 draws of T_{10}^* .

C.2. A 30-minute aggregation diagnostic

Figure 8 gives a small diagnostic for the 30-minute row of Table 5. It is not a size experiment. It is designed to isolate the aggregation mechanism that can make the fixed- k window test reject more often than pooled BNS tests in the FOMC application. Each Monte Carlo draw contains 30 one-minute returns, partitioned into three blocks of length $k = 10$. For a volatility scale $s \geq 1$, the data-generating process is

$$R_i = \begin{cases} Z_i + \eta 1_{\{i=6\}}, & i = 1, \dots, 10, \\ sZ_i, & i = 11, \dots, 30, \end{cases} \quad Z_i \stackrel{iid}{\sim} N(0, 1), \quad \eta = 6.$$

Thus the first block contains a single local price jump, while the remaining two blocks contain only continuous shocks whose volatility level is scaled by s . The fixed- k statistic is the Šidák maximum over the three 10-minute block statistics. BNS- $\hat{H}$ and BNS- $\hat{J}$ are instead computed

once on the pooled 30-return window and use their one-sided 5% lower-tail normal critical value. As s increases, the local fixed- k rejection probability is nearly flat, because the jump block is tested locally and the scale of the other two blocks cancels within their own ratios. The pooled BNS rejection probabilities fall sharply, because the high-volatility non-jump blocks dilute the window-level RV–BV discrepancy and inflate the pooled studentisation. The 30-minute comparison in Table 5 is therefore an aggregation effect, not an implementation anomaly.

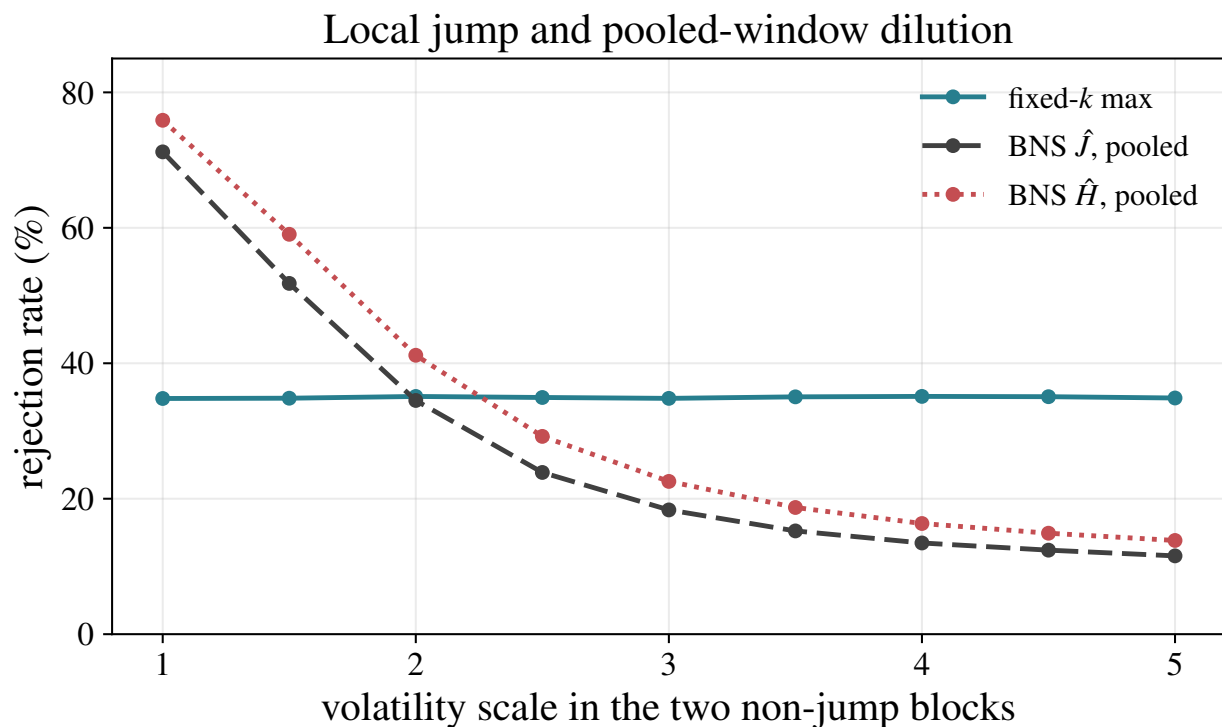


Figure 8: A 30-minute aggregation diagnostic. Each Monte Carlo draw contains 30 Gaussian returns split into three 10-return blocks. The first block contains one jump of size $\eta = 6$; the two remaining blocks have no jumps and continuous-volatility scale s on the horizontal axis. The fixed- k statistic is the Šidák maximum over the three block statistics. BNS- $\hat{H}$ and BNS- $\hat{J}$ are computed once on the pooled 30-return window. Nominal level is 5% and 200,000 replications are used.

C.3. Additional empirical checks

Table 7: Volatility-jump diagnostics for the 30-minute fixed- k -only price-jump cases.

Date	Time	Max block	T_{30m}	BNS- $\hat{J}_{30m}$	$\hat{c}_{1+}/\hat{c}_{1-}$	Vol. jump
2000-06-28	14:15	[-10, -1]	3.00**	0.29	3.11**	Yes
2001-12-11	14:14	[-10, -1]	13.73***	-0.81	6.28***	Yes
2003-05-06	14:13	[-10, -1]	6.30***	1.40	20.67***	Yes
2004-08-10	14:15	[0, +9]	2.65**	-0.75	6.25***	Yes
2006-01-31	14:14	[-10, -1]	4.59***	0.21	5.59***	Yes
2006-08-08	14:15	[-10, -1]	28.99***	-1.22	9.41***	Yes
2007-08-07	14:15	[-10, -1]	11.64***	-1.35*	17.86***	Yes
2008-01-30	14:15	[-10, -1]	9.49***	0.22	11.01***	Yes
2008-03-18	14:15	[-10, -1]	3.00**	-1.20	3.92**	Yes
2008-09-16	14:15	[-10, -1]	10.29***	-1.29*	8.39***	Yes
2009-06-24	14:15	[-10, -1]	4.01***	-0.17	19.44***	Yes
2012-12-12	12:30	[-10, -1]	4.01***	-1.50*	5.81***	Yes
2013-06-19	14:00	[-10, -1]	3.21**	-0.23	16.84***	Yes
2013-12-18	14:00	[-10, -1]	5.95***	0.98	4.58**	Yes
2014-03-19	14:00	[-10, -1]	91.16***	0.97	12.22***	Yes
2014-10-29	14:00	[-10, -1]	7.23***	0.00	4.71***	Yes
2015-01-28	14:00	[-10, -1]	3.19**	-0.55	2.83**	Yes
2015-03-18	14:00	[-10, -1]	4.73***	0.40	2.82**	Yes
2015-04-29	14:00	[-10, -1]	5.99***	-0.36	6.92***	Yes
2015-10-28	14:00	[-10, -1]	2.96**	-1.28	3.62***	Yes
2015-12-16	14:00	[-10, -1]	5.86***	-0.04	3.74***	Yes
2016-04-27	14:00	[-10, -1]	3.56***	-0.10	7.33***	Yes
2016-09-21	14:00	[+10, +19]	2.40**	-1.48*	1.27	No
2017-07-26	14:00	[-10, -1]	7.06***	-1.63*	14.79***	Yes
2019-10-30	14:00	[-10, -1]	3.48***	-0.10	5.88***	Yes

Note: the table reports the 25 FOMC announcements for which the fixed- k window statistic T_{30m} rejects at the 5% Šidák critical value, while pooled BNS- $\hat{J}$ computed on the same 30 one-minute returns does not reject at the one-sided 5% critical value. “Max block” reports the 10-minute block attaining T_{30m} . The BLL volatility-jump statistic is recomputed at $k' = 15$ using two 15-minute pre/post-announcement blocks separated by the 5-minute buffer. Stars denote rejection at the 10%, 5%, and 1% levels, respectively. The critical values are 2.011, 2.390, and 3.464 for T_{30m} ; -1.282, -1.645, and -2.326 for the left-tail BNS- $\hat{J}_{30m}$ test; and 1.972, 2.403, and 3.522 for the right-tail $F_{15,15}$ volatility-jump test. “Vol. jump” equals Yes at the 5% level.

Table 8: SPY 1-min counterpart of Table 5: fixed- k bipower ($k = 10$) vs. BNS- $\hat{H}$ vs BNS- $\hat{J}$ on FOMC announcements.

Testing periods	# log-returns	fixed- k rej.	BNS- $\hat{H}$ rej.	BNS- $\hat{J}$ rej.	Both fixed- k and BNS- $\hat{J}$ rej.
$T_{30m,1}$ block: [-10, -1] min	10	46	75	51	46
$T_{30m,2}$ block: [0, +9] min	10	23	45	27	22
$T_{30m,3}$ block: [+10, +19] min	10	12	34	17	12
T_{30m} Šidák ($m=3$): [-10, +20] min	30	51	45	37	22
FOMC whole days	390	71	76	76	33

Note: three jump tests on three lengths of periods are conducted on 201 weekday FOMC meetings over years 2000–2024 for which SPY 1-min bars are available. The table reports the rejection counts in the 201 FOMC announcements of the fixed- k bipower jump test and the BNS- $\hat{H}$ & - $\hat{J}$ tests under windows of different lengths, all at significance level $\alpha = 5\%$. For the 30-minute and whole-day rows, BNS- $\hat{H}$ and BNS- $\hat{J}$ are computed once on the pooled returns over the corresponding horizon, whereas fixed- k entries are Šidák maxima over non-overlapping 10-minute blocks. For testing jumps on FOMC whole days, BNS- $\hat{H}$ and BNS- $\hat{J}$ reject on the exact same 76 days.

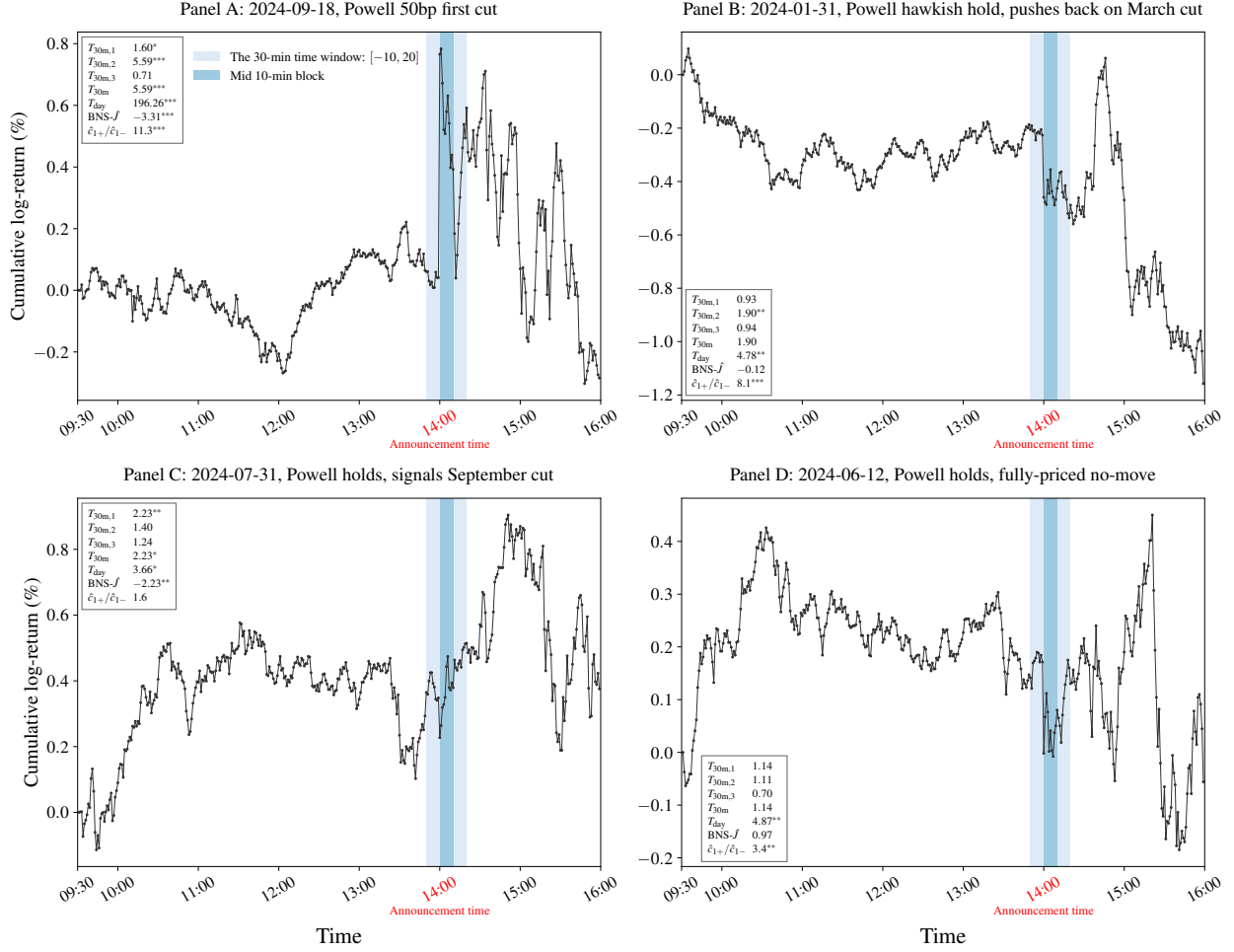


Figure 9: SPY 1-min counterpart of Figure 5: cumulative 1-min log-return from the 09:30 ET open on four FOMC meetings, using the SPY ETF instead of S&P 500 E-mini futures. Each panel’s annotation box reports seven statistics: (i) the fixed- k bipower price-jump test with $k=10$ at three nested levels—the three contiguous 10-min blocks tiling $[-10, +20]$ window, $T_{30m,1}$, $T_{30m,2}$, $T_{30m,3}$; their Šidák max over the 30-min window, T_{30m} ; and the Šidák max over the whole-day $m=39$ blocks, T_{day} . (ii) the day-level BNS- $\hat{J}$ adjusted ratio jump test. (iii) the fixed- k volatility jump test $\hat{c}_{1+}/\hat{c}_{1-}$. Superscripts denote significance levels: * at 10%, ** at 5%, *** at 1%.

Table 9: SPY 1-min counterpart of Table 6: cross-classification of FOMC meetings: price jump $\times$ vol jump.

	Vol jump: Yes	Vol jump: No	Total
Price jump: Yes	44	7	51
Price jump: No	126	24	150
Total	170	31	201

Note: the 2-by-2 price-volatility jump classification table on $N = 201$ weekday FOMC meetings over the years 2000–2024 for the SPY ETF. Price-jump rejection from the fixed- k bipower test over the $[-10, +20]$ -minute window with $k = 10$; vol-jump rejection from the $\hat{c}_{1+}/\hat{c}_{1-}$ test in [Bollerslev, Li, and Liao \(2021\)](#) on two 15-minute pre/post-announcement blocks separated by a 5-minute buffer around the announcement. Significance level is set to be 5%.

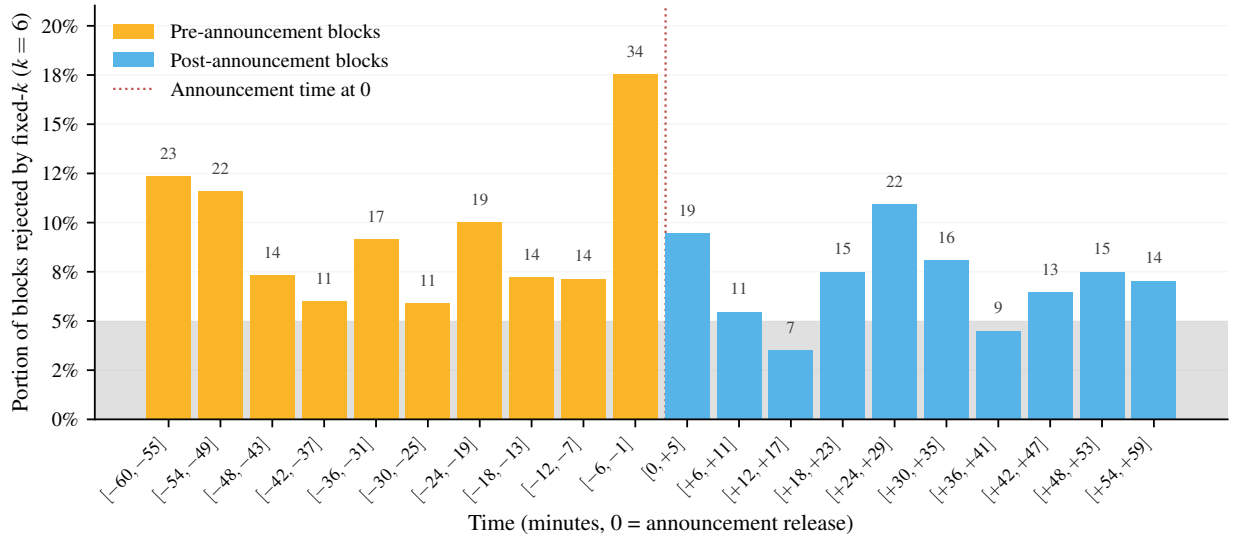


Figure 10: SPY 1-min counterpart of Figure 6: fixed- k bipower jump-test rejection rates on twenty non-overlapping 6-minute blocks tiling the $[-60, +60]$ -minute window around FOMC announcements, with $k=6$ 1-minute returns per block and the single-block 5% critical value $q_6^*(0.95)$. Sample: $N = 201$ weekday FOMC meetings (198 scheduled + 3 regular-hours inter-meeting actions, 2000–2024, SPY ETF 1-minute bars). Each bar gives the share of meetings on which the block's $T_{n,j}$ exceeds the critical value; the integer above each bar is the rejection count on the days for which the block fully fits within the available regular-trading-hours coverage. The light-gray band marks the nominal 5% size; the red dotted vertical line marks the announcement instant (0 on the time axis).

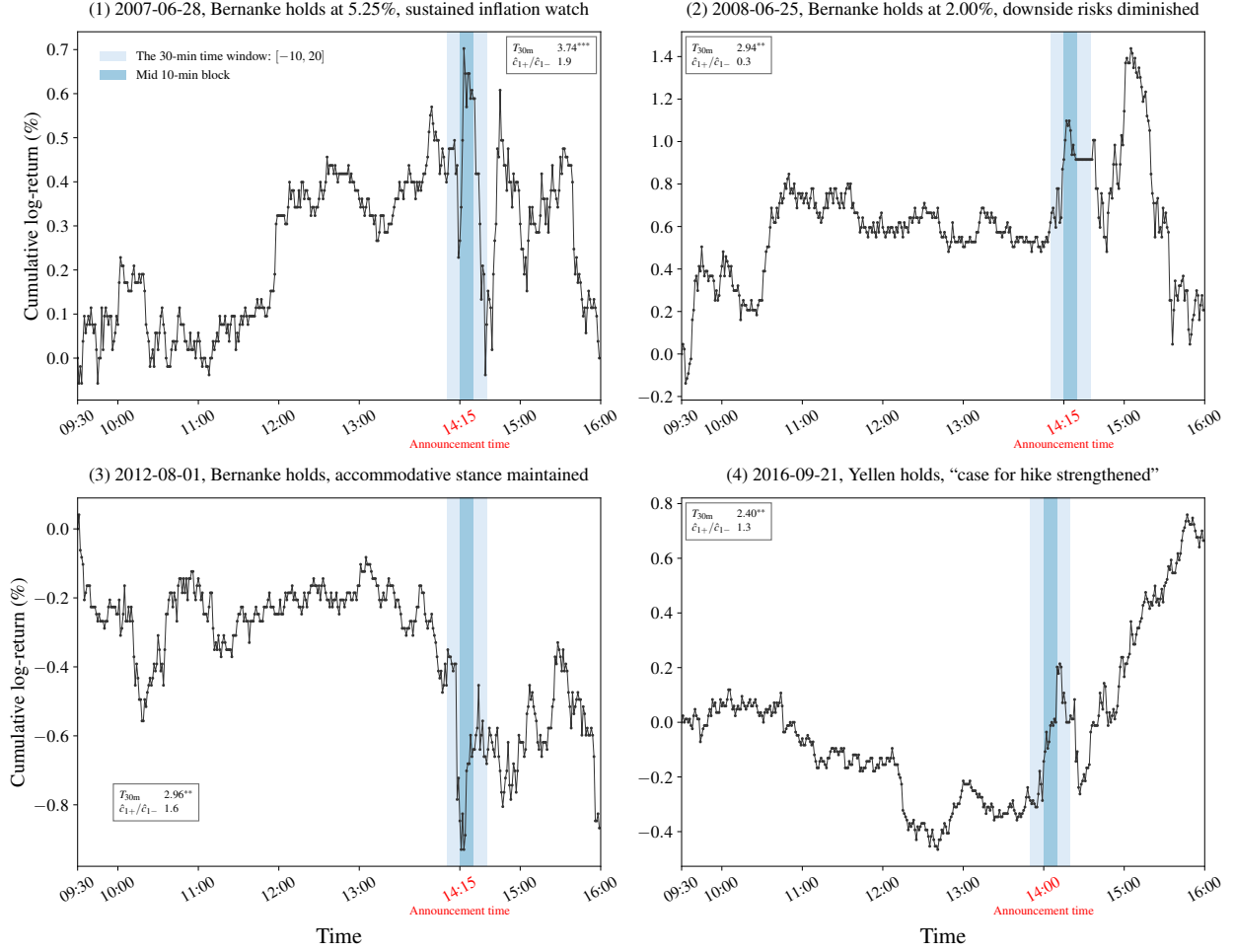


Figure 11: The four FOMC meetings populating the price-jump=Yes, vol-jump=No cell of Table 6: the fixed- k bipower window-Šidák price-jump test at $k = 10$ rejects, but the BLL fixed- k' $F_{15,15}$ volatility-jump test does not reject. Cumulative 1-min log-return (%) from the 09:30 ET open follows the same convention as Figure 5; each panel's annotation reports the window-Šidák max T_{30m} and the BLL F-statistic $\hat{c}_{1+}/\hat{c}_{1-}$ (computed at $k' = 15$, asymptotic $F_{15,15}$ -based 5%-level critical value 2.403); the full seven-statistic format used in Figure 5 is dropped here to focus on the two headline tests of Table 6.

Appendix D. Additional tables for critical values

Table 10: Extended quantile table for T_k^* , $k \in \{4, \dots, 50\}$.

k	LB	med.	$q_k^*(.90)$	$q_k^*(.95)$	$q_k^*(.99)$	k	LB	med.	$q_k^*(.90)$	$q_k^*(.95)$	$q_k^*(.99)$
4	0.590	0.958	2.421	3.606	8.919	28	0.617	0.992	1.246	1.348	1.593
5	0.588	0.970	2.052	2.862	6.243	29	0.618	0.992	1.241	1.342	1.579
6	0.589	0.972	1.860	2.447	4.586	30	0.619	0.992	1.234	1.331	1.564
7	0.591	0.974	1.728	2.194	3.761	31	0.619	0.992	1.230	1.324	1.544
8	0.593	0.978	1.639	2.021	3.249	32	0.620	0.993	1.226	1.318	1.533
9	0.595	0.980	1.579	1.913	2.927	33	0.620	0.993	1.221	1.310	1.520
10	0.597	0.980	1.525	1.813	2.671	34	0.620	0.993	1.216	1.303	1.503
11	0.599	0.981	1.485	1.747	2.504	35	0.621	0.993	1.213	1.298	1.494
12	0.601	0.983	1.453	1.688	2.351	36	0.621	0.994	1.209	1.294	1.486
13	0.603	0.984	1.424	1.636	2.234	37	0.622	0.994	1.205	1.287	1.477
14	0.604	0.985	1.402	1.599	2.131	38	0.622	0.994	1.202	1.281	1.467
15	0.606	0.986	1.379	1.565	2.059	39	0.622	0.994	1.200	1.278	1.456
16	0.607	0.986	1.362	1.535	1.993	40	0.623	0.994	1.196	1.272	1.448
17	0.608	0.987	1.347	1.510	1.934	41	0.623	0.994	1.193	1.268	1.439
18	0.610	0.988	1.333	1.486	1.885	42	0.623	0.995	1.190	1.264	1.434
19	0.611	0.988	1.319	1.464	1.833	43	0.623	0.994	1.188	1.260	1.426
20	0.612	0.989	1.309	1.448	1.800	44	0.624	0.995	1.185	1.256	1.417
21	0.613	0.990	1.299	1.432	1.760	45	0.624	0.995	1.182	1.252	1.409
22	0.613	0.990	1.290	1.418	1.733	46	0.624	0.995	1.180	1.249	1.404
23	0.614	0.991	1.281	1.404	1.703	47	0.624	0.995	1.177	1.245	1.397
24	0.615	0.990	1.271	1.390	1.677	48	0.625	0.995	1.174	1.242	1.393
25	0.616	0.991	1.265	1.378	1.654	49	0.625	0.995	1.173	1.239	1.387
26	0.616	0.991	1.258	1.368	1.634	50	0.625	0.995	1.170	1.235	1.382
27	0.617	0.991	1.253	1.359	1.616						

Note: Extension of Table 1 to every block size $k \in \{4, \dots, 50\}$. “LB” is the sharp support endpoint $L_k = \mu_1^2(k-1)/[k \cos(\pi/(k+1))]$ from Proposition 2; “med.” is the median of T_k^* . The calculation is based on 500,000 Monte Carlo trials.

Table 11: Šidák-corrected critical values for the fixed- k bipower jump test on multiple blocks.

$k \backslash m$	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39
Panel: $\alpha = 10\%$																																						
4	3.56	4.83	5.77	6.49	7.09	7.59	8.03	8.42	8.76	9.21	9.74	10.22	10.67	11.08	11.47	11.84	12.19	12.51	12.82	13.12	13.69	14.02	14.37	14.68	15.01	15.31	15.63	15.97	16.29	16.59	16.88	17.18	17.45	17.76	18.05	18.35	18.63	18.91
5	2.83	3.64	4.24	4.70	5.08	5.40	5.68	5.92	6.14	6.42	6.73	7.02	7.29	7.54	7.78	8.00	8.21	8.40	8.59	8.77	8.92	9.12	9.31	9.50	9.69	9.87	10.04	10.22	10.39	10.57	10.74	10.90	11.06	11.22	11.38	11.53	11.67	11.83
6	2.43	2.94	3.32	3.61	3.85	4.05	4.23	4.38	4.52	4.68	4.86	5.02	5.16	5.30	5.43	5.55	5.67	5.77	5.88	5.97	6.12	6.23	6.33	6.43	6.52	6.61	6.70	6.79	6.88	6.96	7.05	7.13	7.21	7.30	7.38	7.46	7.54	7.62
7	2.18	2.55	2.83	3.05	3.22	3.37	3.50	3.61	3.72	3.83	3.95	4.06	4.16	4.26	4.35	4.44	4.52	4.59	4.66	4.73	4.83	4.91	4.98	5.04	5.11	5.18	5.24	5.30	5.36	5.42	5.47	5.53	5.58	5.63	5.68	5.74	5.79	5.84
8	2.01	2.30	2.52	2.69	2.83	2.94	3.04	3.13	3.21	3.30	3.39	3.48	3.56	3.63	3.70	3.76	3.82	3.88	3.94	3.99	4.06	4.11	4.16	4.21	4.26	4.31	4.35	4.40	4.44	4.48	4.53	4.57	4.60	4.64	4.68	4.71	4.75	4.79
9	1.90	2.15	2.32	2.46	2.58	2.67	2.76	2.83	2.90	2.97	3.04	3.10	3.17	3.22	3.28	3.33	3.37	3.42	3.46	3.50	3.54	3.57	3.61	3.65	3.69	3.73	3.76	3.80	3.83	3.86	3.89	3.92	3.95	3.98	4.02	4.05	4.08	4.10
10	1.80	2.01	2.16	2.28	2.38	2.46	2.53	2.59	2.65	2.70	2.76	2.82	2.87	2.91	2.95	3.00	3.03	3.07	3.10	3.14	3.18	3.22	3.25	3.28	3.31	3.34	3.37	3.40	3.43	3.45	3.48	3.50	3.53	3.55	3.57	3.60	3.62	3.64
11	1.74	1.92	2.05	2.16	2.24	2.32	2.38	2.43	2.48	2.53	2.58	2.63	2.67	2.71	2.74	2.78	2.81	2.84	2.87	2.90	2.92	2.95	2.98	3.00	3.03	3.05	3.08	3.10	3.12	3.14	3.17	3.19	3.21	3.23	3.25	3.27	3.29	3.30
12	1.68	1.84	1.96	2.05	2.12	2.19	2.24	2.29	2.33	2.38	2.42	2.46	2.49	2.53	2.56	2.59	2.62	2.65	2.67	2.70	2.72	2.74	2.77	2.79	2.81	2.83	2.85	2.87	2.89	2.91	2.93	2.95	2.97	2.98	3.00	3.01	3.03	3.05
13	1.63	1.77	1.88	1.96	2.03	2.08	2.13	2.18	2.22	2.26	2.29	2.33	2.36	2.40	2.42	2.45	2.48	2.50	2.53	2.55	2.56	2.58	2.60	2.62	2.64	2.66	2.68	2.70	2.71	2.73	2.75	2.76	2.78	2.79	2.80	2.82	2.83	2.85
14	1.59	1.72	1.82	1.89	1.95	2.00	2.04	2.08	2.12	2.15	2.18	2.21	2.24	2.27	2.30	2.32	2.34	2.36	2.38	2.40	2.43	2.45	2.46	2.48	2.50	2.51	2.53	2.55	2.56	2.58	2.59	2.60	2.62	2.63	2.64	2.65	2.67	2.68
15	1.56	1.68	1.77	1.83	1.89	1.94	1.98	2.01	2.04	2.08	2.11	2.13	2.16	2.18	2.21	2.23	2.25	2.27	2.28	2.30	2.32	2.34	2.35	2.37	2.38	2.40	2.41	2.42	2.44	2.45	2.46	2.47	2.49	2.50	2.51	2.52	2.53	2.54
Panel: $\alpha = 5\%$																																						
4	5.85	7.18	8.12	8.85	9.90	10.83	11.64	12.35	12.99	13.88	14.56	15.22	15.89	16.53	17.14	17.74	18.34	18.92	19.50	20.07	20.63	21.11	21.61	22.08	22.57	22.99	23.48	23.96	24.42	24.89	25.30	25.71	26.10	26.56	27.04	27.47	27.88	28.24
5	4.29	5.13	5.73	6.20	6.83	7.39	7.87	8.30	8.69	9.03	9.44	9.82	10.18	10.54	10.88	11.21	11.52	11.83	12.13	12.44	12.73	13.01	13.31	13.60	13.87	14.14	14.39	14.65	14.90	15.14	15.38	15.61	15.84	16.06	16.30	16.53	16.77	16.98
6	3.35	3.88	4.26	4.56	4.91	5.22	5.48	5.72	5.93	6.19	6.39	6.58	6.76	6.95	7.12	7.29	7.46	7.62	7.76	7.91	8.06	8.19	8.31	8.44	8.57	8.69	8.81	8.93	9.04	9.16	9.29	9.40	9.51	9.61	9.72	9.83	9.93	10.03
7	2.86	3.25	3.53	3.74	3.99	4.20	4.39	4.55	4.70	4.88	5.02	5.16	5.28	5.41	5.52	5.63	5.73	5.84	5.95	6.04	6.14	6.23	6.32	6.41	6.49	6.57	6.65	6.74	6.82	6.90	6.97	7.04	7.12	7.18	7.26	7.32	7.39	7.45
8	2.54	2.85	3.06	3.23	3.42	3.58	3.73	3.85	3.96	4.09	4.20	4.29	4.38	4.48	4.56	4.64	4.71	4.79	4.86	4.93	4.99	5.06	5.12	5.18	5.24	5.29	5.35	5.40	5.46	5.51	5.57	5.62	5.67	5.71	5.76	5.80	5.85	5.90
9	2.34	2.59	2.77	2.91	3.06	3.19	3.30	3.40	3.48	3.56	3.64	3.72	3.79	3.85	3.92	3.98	4.05	4.11	4.16	4.22	4.27	4.32	4.37	4.42	4.46	4.50	4.55	4.59	4.63	4.66	4.70	4.74	4.77	4.81	4.84	4.88	4.91	4.94
10	2.18	2.39	2.54	2.66	2.78	2.88	2.97	3.05	3.12	3.20	3.27	3.33	3.39	3.45	3.50	3.55	3.60	3.64	3.68	3.73	3.77	3.81	3.84	3.88	3.91	3.95	3.98	4.02	4.05	4.08	4.11	4.14	4.16	4.19	4.21	4.24	4.26	4.29
11	2.07	2.26	2.39	2.49	2.60	2.68	2.76	2.82	2.88	2.94	3.00	3.05	3.09	3.14	3.18	3.23	3.27	3.31	3.34	3.38	3.41	3.44	3.47	3.50	3.53	3.56	3.59	3.62	3.65	3.67	3.70	3.72	3.75	3.77	3.79	3.81	3.84	3.86
12	1.97	2.13	2.25	2.34	2.43	2.51	2.57	2.63	2.69	2.73	2.78	2.83	2.87	2.91	2.95	2.98	3.01	3.05	3.08	3.11	3.13	3.16	3.19	3.21	3.24	3.26	3.28	3.31	3.33	3.35	3.37	3.39	3.41	3.43	3.45	3.47	3.49	3.50
13	1.89	2.04	2.14	2.23	2.31	2.38	2.44	2.49	2.54	2.57	2.62	2.66	2.69	2.73	2.76	2.79	2.82	2.85	2.87	2.90	2.92	2.95	2.97	2.99	3.01	3.04	3.06	3.08	3.10	3.12	3.14	3.15	3.17	3.19	3.20	3.22	3.23	3.25
14	1.82	1.96	2.05	2.12	2.19	2.25	2.31	2.35	2.39	2.44	2.48	2.51	2.54	2.57	2.60	2.63	2.65	2.68	2.70	2.73	2.75	2.77	2.79	2.81	2.83	2.85	2.87	2.88	2.90	2.92	2.93	2.95	2.96	2.98	2.99	3.00	3.02	3.03
15	1.77	1.90	1.98	2.05	2.11	2.17	2.21	2.26	2.29	2.33	2.36	2.39	2.42	2.45	2.47	2.50	2.52	2.54	2.56	2.58	2.60	2.62	2.64	2.66	2.67	2.69	2.71	2.72	2.74	2.75	2.76	2.78	2.79	2.80	2.81	2.82	2.84	2.85

Continued on next page.

Table 11 (continued).

$k \backslash m$	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39
Panel: $\alpha = 1\%$																																						
4	13.11	16.72	19.73	22.35	24.70	26.89	28.92	30.94	32.82	34.53	36.13	37.82	39.28	40.62	42.08	43.53	45.01	46.52	47.79	49.22	50.62	51.75	53.02	54.33	55.65	56.78	58.03	59.18	60.30	61.48	62.47	63.42	64.50	65.43	66.28	67.05	67.91	69.26
5	8.76	10.64	12.27	13.74	15.04	16.24	17.34	18.41	19.43	20.35	21.21	22.06	22.89	23.66	24.41	25.26	25.95	26.64	27.39	28.04	28.76	29.33	29.98	30.68	31.21	31.77	32.18	32.78	33.29	33.75	34.38	35.01	35.43	35.84	36.31	36.83	37.33	37.77
6	5.97	7.00	7.83	8.50	9.11	9.69	10.21	10.68	11.10	11.53	11.94	12.34	12.70	13.04	13.33	13.63	13.92	14.19	14.45	14.74	15.02	15.27	15.53	15.77	16.00	16.27	16.47	16.70	16.90	17.07	17.27	17.44	17.66	17.83	18.10	18.34	18.47	18.69
7	4.73	5.44	5.99	6.45	6.87	7.24	7.57	7.87	8.15	8.41	8.66	8.88	9.11	9.32	9.53	9.74	9.93	10.10	10.26	10.41	10.57	10.72	10.92	11.05	11.18	11.31	11.48	11.62	11.77	11.91	12.02	12.15	12.26	12.43	12.53	12.65	12.77	12.89
8	3.99	4.50	4.89	5.21	5.49	5.74	5.97	6.17	6.36	6.52	6.70	6.86	7.01	7.15	7.27	7.38	7.51	7.63	7.72	7.84	7.95	8.07	8.16	8.25	8.35	8.43	8.52	8.59	8.69	8.79	8.87	8.96	9.03	9.10	9.17	9.25	9.34	9.43
9	3.50	3.88	4.19	4.44	4.65	4.83	5.00	5.16	5.29	5.42	5.54	5.66	5.77	5.88	5.97	6.06	6.16	6.24	6.32	6.41	6.47	6.54	6.61	6.69	6.75	6.83	6.90	6.97	7.03	7.08	7.15	7.21	7.27	7.32	7.36	7.41	7.47	7.52
10	3.14	3.46	3.70	3.90	4.07	4.21	4.34	4.46	4.57	4.68	4.76	4.85	4.93	5.01	5.07	5.14	5.22	5.29	5.36	5.42	5.47	5.53	5.58	5.62	5.67	5.72	5.77	5.83	5.88	5.92	5.96	6.01	6.05	6.08	6.12	6.16	6.20	6.24
11	2.90	3.15	3.36	3.52	3.66	3.78	3.90	4.00	4.08	4.17	4.24	4.31	4.39	4.44	4.51	4.56	4.62	4.68	4.73	4.80	4.85	4.89	4.94	4.98	5.02	5.07	5.09	5.13	5.18	5.22	5.25	5.27	5.31	5.34	5.38	5.41	5.44	5.47
12	2.70	2.92	3.09	3.23	3.34	3.44	3.53	3.61	3.69	3.76	3.83	3.89	3.94	3.99	4.05	4.10	4.14	4.18	4.23	4.27	4.30	4.34	4.38	4.42	4.46	4.49	4.53	4.56	4.59	4.63	4.65	4.68	4.70	4.73	4.74	4.77	4.80	4.82
13	2.55	2.74	2.88	3.00	3.11	3.20	3.28	3.35	3.42	3.47	3.53	3.58	3.62	3.67	3.71	3.75	3.79	3.82	3.86	3.89	3.92	3.95	3.98	4.01	4.03	4.05	4.08	4.11	4.13	4.15	4.17	4.19	4.21	4.23	4.24	4.26	4.28	4.29
14	2.40	2.58	2.71	2.82	2.91	2.99	3.05	3.12	3.18	3.23	3.28	3.32	3.37	3.41	3.44	3.48	3.51	3.54	3.57	3.60	3.63	3.66	3.68	3.71	3.74	3.76	3.78	3.81	3.83	3.85	3.88	3.89	3.91	3.93	3.95	3.96	3.98	4.00
15	2.30	2.46	2.57	2.66	2.74	2.81	2.87	2.92	2.97	3.02	3.06	3.10	3.14	3.17	3.21	3.23	3.26	3.29	3.31	3.34	3.36	3.39	3.41	3.43	3.45	3.47	3.49	3.51	3.53	3.56	3.58	3.59	3.61	3.63	3.64	3.66	3.67	3.69

Note: Šidák-corrected critical values $q_k^*((1 - \alpha)^{1/m})$ for the multi-block fixed- k bipower jump test $T_{n,j}^{\max}$. Rows: block size k . Columns: number of Šidák blocks m . Three panels for nominal levels $\alpha \in \{10\%, 5\%, 1\%\}$. The cell at $(k, m, \alpha) = (10, 39, 5\%)$ equals 4.29 — the day-level critical value used throughout the empirical application of Section 7 at 1-min sampling.